DOMAIN 2 STUDY QUESTIONS

Question 1

Question: Preliminary Hazard Analysis is a system safety technique that:

- a. Allows the results to be presented in the form of a table or logic diagram
- b. Allows the results to be presented in the form of "and" gates
- c. Starts with an undesired event within the system
- d. Employs Boolean logic

Correct Answer: Select (a)
Explanation: PHA is a qualitative system safety technique that allows its results to be presented in the form of a table or logic diagram. See the description in the text.

In either format, potential hazards that pose a high risk are identified, along with their cause and consequences. In addition, for each hazard identified, a preliminary means of control is prescribed in the analysis.

Question 2

Question: The main objective of a Hazard and Operability Study (HAZOP) is to:

- a. Identify deviations from the design intent with their causes and consequences.
- b. Develop "guide words" that would best describe system hazard analysis.
- c. Satisfy OSHA requirement of process safety management standard (29 CFR 1910.119).
- d. Apply guide words to safety instruments.

Correct Answer: Select (a)
Explanation: The main purpose of HAZOP is to determine how a process can deviate from its design intent and what the causes and consequences of these deviations would be. It must be pointed out that the main objective in a HAZOP study is problem identification not problem solution. However, if the solution to a problem becomes obvious during the course of study, the solution is recorded.

Question 3

Question: Fault Tree Analysis (FTA):

- a. Is a forward analysis
- b. Is a backward analysis
- c. Employs Boolean logic
- d. Only (b) and (c) above

Correct Answer: Select (d)
Explanation: FTA is a backward analysis. This means that the accident or system hazard is traced backward to all its possible causes.

Three steps are required to conduct a fault tree analysis thoroughly and accurately. First, the undesired event, or top event, is defined. In the second step, investigators develop a thorough understanding of the system to be analyzed. This can be accomplished by studying design drawings, equipment specifications, literature, and operation procedures, as well as any other source information that may be available. The third step is constructing the fault tree.

Question 4

Question: In FTA, the event "source of ignition" as one of the necessary components of combustion must be connected to events "open flame," "static sparks," and "electrical sparks" through which of the following gates?

- a. AND
- b. SOMETIMES
- c. OR
- d. Only (a) and (c) above

Correct Answer: Select (c)
Explanation: AND gates are used when the existence of two or more simultaneous conditions is required. OR gates are used when any of the events leads to an upper event in the tree. In this case, any of the three events mentioned lead to the event "source of ignition."

Question 5

Question: In Fault Tree Analysis of a fire situation, air, fuel, source of ignition, and chemical chain reaction are required in order to have a fire. These events must be connected to the TOP event "fire" through an:

- a. OR gate
- b. EXTERNAL gate
- c. UNDESIRED EVENT gate
- d. AND gate

Correct Answer: Select (d)
Explanation: "AND" gates are used when the existence of two or more simultaneous conditions is required. In this case all four events are required simultaneously to lead to the event "fire."

Question 6

Question: If 5 items fail out of 10,000 tested under specified conditions, the reliability of the item is:

- a. 99.5%
- b. 99.95%
- c. 95.0%
- d. 90.0%

Correct Answer: Select (b)
Explanation: $\text{Reliability} = 1 - \frac{\text{number of failures}}{\text{total number of items exposed}}$
$R = 1 - \frac{5}{10,000} = 99.95\%$

Question 7

Question: Three components each having a reliability of 0.99, 0.95, and 0.98 are arranged in series. The system reliability is:

- a. 0.99
- b. 0.95

- c. 0.92
- d. 0.85

Correct Answer: Select (c)
Explanation: $R_{(system)} = R_1 \times R_2 \times R_3$
$R_{(system)} = 0.99 \times 0.95 \times 0.98 = 92%$

Question 8

Question: Two components arranged in parallel have reliabilities of 0.999 and 0.998 respectively. The system reliability is:

- a. 99.99%
- b. 99.80%
- c. 99.70%
- d. 99.00%

Correct Answer: Select (a)
Explanation: $R_{system} = 1 - (1 - R_1)(1 - R_2)$
$R_{system} = 1 - (1 - 0.999)(1 - 0.998) = 99.99%$

Question 9

Question: Which of the following statements is true regarding a HAZOP study?

- a. Identify deviations from design intent
- b. Uses guidewords
- c. Satisfies OSHA PSM requirements
- d. All of the above

Correct Answer: Select (d)
Explanation: Statements in (a), (b), and (c) are all true about a HAZOP study. Fault Tree Analysis uses Boolean logic.

Question 10

Question: Which of the following statements is NOT true in regards to FMECA?

- a. It uses guidewords to identify deviation of process parameters
- b. It is an inductive analysis
- c. Equipment and system failures and resulting effects are determined
- d. Quantitative data can later be applied to establish criticality ranking

Correct Answer: Select (a)
Explanation: Statements in (b), (c), and (d) are true about FMECA. HAZOP uses guidewords to identify deviation of process parameters from their design intent.

Question 11

Question: In Fault Tree Analysis, the pentagon symbol represents a(n):

- a. Basic fault
- b. Undeveloped event
- c. Initiating event
- d. External event

Correct Answer: Select (d)
Explanation: External events in Fault Tree Analysis are those events that one has little control over. Examples are the existence of air, difference in water temperature between summer and winter time, etc.

Basic faults are the primary cause of the top event. Basic faults represent a malfunction of equipment that occurs in the environment in which the equipment was intended to operate. Undeveloped events are occurrences that are not further addressed, either for lack of necessary information or for other reasons. Each branch of the fault tree should eventually end up in either a basic fault or in an undeveloped event.

Question 12

Question: If a device has a failure rate of 2×10^{-6} failures/hour, its reliability in an operating period of 500 hours is:

- a. 0.999
- b. 0.989
- c. 0.977
- d. 0.960

Correct Answer: Select (a)
Explanation: $R_{500} = \exp(-2 \times 10^{-6} \times 500) = e^{-0.001} = 0.999$ or the probability of failure 0.001.

Question 13

Question: The system safety technique which is a forward analysis and begins with an initiating event and proceeds forward to find possible consequences is called:

- a. Event Tree Analysis
- b. Fault Tree Analysis
- c. HAZOP
- d. Cause-Consequence Analysis

Correct Answer: Select (a)
Explanation: Event tree analysis (ETA) is an analysis beginning with an initiating event and proceeding forward to find possible consequences resulting from that event. The course of events is determined by the success or failure of various safety functions (controls) as the accident progresses.

Question 14

Question: The system safety technique that is primarily devoted to the analysis of human errors in the workplace is called:

- a. HEA
- b. THERP
- c. PERT
- d. HFE

Correct Answer: Select (b)
Explanation: The Technique for Human Error Rate Prediction (THERP), is a system safety technique which can identify and analyze human errors in conducting a given task. THERP provides the means to reduce the probability of errors occurring and therefore can greatly improve the overall safety within a system.

Question 15

Question: The system safety technique that evaluates the success or failure of safety functions is called:

- a. System Safety Analysis
- b. Failure Mode Effect Analysis
- c. Event Tree Analysis
- d. Fault Tree Analysis

Correct Answer: Select (c)
Explanation: ETA begins with an initiating event and analyzes both the success and failure of safety functions that are designed to control the initiating event.

Question 16

Question: A backward system safety analysis technique that visually demonstrates the interrelationships among equipment failure, human error, and environmental factors that can result in occurrence of an accident is called:

- a. Event Tree Analysis
- b. Fault Tree Analysis
- c. THERP
- d. EHE analysis

Correct Answer: Select (b)
Explanation: This is a good description of Fault Tree Analysis.

Question 17

Question: The system safety technique that combines the forward thinking features of Event Tree Analysis with the reverse thinking features of Fault Tree Analysis is called:

- a. Cause-Consequence Analysis
- b. HAZOP
- c. PHA
- d. Job Safety Analysis

Correct Answer: Select (a)
Explanation: CCA results in a diagram that displays the relationships between accident sequences and their basic causes.

Question 18

Question: FMECA is best described by which of the following statements?

- a. A technique to identify problems associated with potential hazards and deviation of plant operation from design specifications.
- b. A systematic method by which equipment and system failures and the resulting effects of these failures are determined.
- c. A technique that visually demonstrates the interrelationship among equipment failure, human error, and environmental factors that can result in an accident.
- d. A forward analysis beginning with an initiating event and proceeding forward to find possible consequences resulting from that event.

Correct Answer: Select (b)
Explanation: FMECA is best described by statement (b). Also, FMECA is an inductive analysis: that is, possible events are studied, but not the reasons for their occurrences.

Question 19

Question: A system has three components. Each component works independently of the others. If the failure probability of each component is F, what is the probability that all three components fail at the same time?

- a. 3F
- b. 6F
- c. 5F
- d. F^3

Correct Answer: Select (d)
Explanation: Since each component can fail independently, the probabilities are multiplied: $F \times F \times F = F^3$.

Question 20

Question: In Fault Tree Analysis, the probability of a top event directly connected to two events with probabilities of P and Q through an OR gate is equal to:

- a. $P - Q + (PQ)$
- b. $P + Q + (PQ)$
- c. $P + Q - (PQ)$
- d. $P + PQ - (p)$

Correct Answer: Select (c)
Explanation: For example, if a top event is connected to two events with probabilities of 0.2 and 0.3 through an OR gate, the probability of top event is: $0.2 + 0.3 - (0.2 \times 0.3) = 0.44$ or $44%$.

Question 21

Question: Which of the following is an accepted probabilistic risk assessment technique that analyzes a system in the context of its environment and operation to find all possible ways that the system can fail?

- a. Fault tree analysis
- b. Event tree analysis
- c. Safety and environmental analysis
- d. Hazard and operability analysis

Correct Answer: Select (a)
Explanation: Fault tree analysis is a deductive method that starts with a top event and determines all possible causes of the top event. Furthermore, fault tree analysis allows the calculation of the probability of the top event.

Question 22

Question: Which of the following is the correct statement of the Distributive Law?

- a. $A \cdot (B \cdot C) = (A \cdot B) + (A \cdot C)$
- b. $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$
- c. $(A \cdot B) + C = (A \cdot C) + (A \cdot B)$
- d. $(A + C) - B = (A \cdot B) - (B \cdot C)$

Correct Answer: Select (b)
Explanation: The Boolean expression states that the union of one subset with two others is equal to the union of their intersections.

Question 23

Question: A pump in a processing facility has a failure rate equal to 10^{-5} failures per hour. An identical spare pump is installed that will automatically start upon failure of the main pump. The probability that this system will fail in 50,000 hours of operation is:

- a. 42.3 percent
- b. 15.5 percent
- c. 10.4 percent
- d. 8.5 percent

Correct Answer: Select (b)
Explanation: In this case we are dealing with a parallel system. In other words, both pumps must fail for the system to fail. Parallel reliability can be calculated from:
$R_{(t)system} = 1 - (1 - R_{1(t)})(1 - R_{2(t)}) \dots$
In this case:
$R_{1(t)} = R_{2(t)} = e^{-10^{-5}(50,000)} = 0.606$
$R_{system} = 1 - (1 - 0.606)(1 - 0.606) = 0.845$
$P_f = 1 - 0.845 = 0.155$ or 15.5 percent.

Question 24

Question: A device with a failure rate of 10^{-6} failures per hour has a Mean Time Between Failures (MTBF) equal to:

- a. 10^4 hours
- b. 10^2 hours
- c. 10^6 hours
- d. 10^{12} hours

Correct Answer: Select (c)
Explanation: $\text{MTBF} = \frac{1}{\lambda} = \frac{1}{10^{-6}} = 10^6\text{ hours}$.

Question 25

Question: A pump feeds cooling water to the jacket around a reactor in which an exothermic chemical reaction takes place. The pump manufacturer's test results indicate a failure rate of 1×10^{-5} failures per hour. The probability that this pump will fail in 50,000 hours of operation is:

- a. 39.4 percent
- b. 21.4 percent
- c. 12.5 percent
- d. 18.7 percent

Correct Answer: Select (a)
Explanation: The reliability in t hours of operation is:
$R_{(t)} = e^{-\lambda t}$ where λ is the failure rate
$R_{(50,000)} = e^{-10^{-5}(50,000)} = 0.606$
Probability of failure is $1 - \text{reliability}$:
$P_f = 1 - R_{(t)} = 1 - 0.606 = 0.394$ or 39.4 percent.

Question 26

Question: The following initiating event probabilities were identified as a result of an FTA analysis on a system:

- Event A: 10^{-8}
- Event B: 10^{-9}
- Event C: 10^{-8}
- Event D: 10^{-8}

The minimal cut sets are: A, B and C, D. The probability of the top event is:

- a. 1×10^{-24}
- b. 11×10^{-17}
- c. 1×10^{-17}
- d. 11×10^{-24}

Correct Answer: Select (b)
Explanation: First we multiply the probabilities within each set:
Set A,B: $10^{-8} \times 10^{-9} = 10^{-17}$
Set C,D: $10^{-8} \times 10^{-8} = 10^{-16} = 10 \times 10^{-17}$

Then we add the probability of all sets to get the probability of top event:
$10^{-17} + 10 \times 10^{-17} = 11 \times 10^{-17}$.

Question 27

Question: A light bulb manufacturer claims a mean time between failures (MTBF) for one of their brands of bulbs to be 10,000 hours. What is the reliability of this type of bulb for 6,000 hours of operation?

- a. 34%
- b. 67%
- c. 45%
- d. 55%

Correct Answer: Select (d)
Explanation: $R_t = e^{-\lambda t}$ where R_t is the reliability in time t, λ is the hazard rate, t is time, $\text{MTBF} = \frac{1}{\lambda}$.
For the problem at hand:
$R_t = e^{\frac{-t}{\text{MTBF}}} = e^{\frac{-6000}{10000}} = 0.55$ or $55%$.
This means that the probability of the bulb operating for 6,000 hours without failure is 55%. The probability that the bulb fails is $P_f = 1 - R_t = 1 - 0.55 = 0.45$ or $45%$.

Question 28

Question: The probabilities of two events connected to the top event through an AND gate are P and Q respectively. What is the probability of the top event?

- a. $P + Q - PQ$
- b. PQ
- c. PQ^2
- d. $P - Q$

Correct Answer: Select (b)
Explanation: For an AND gate, the probability of the output event is the product of the probabilities of the input events (PQ).

Question 29

Question: Which of the following is a correct Boolean expression?

- a. $A + A = A$
- b. $A + 0 = A$
- c. $A \cdot A = A$
- d. All of the above

Correct Answer: Select (d)
Explanation: All of the listed identity laws are valid Boolean algebra expressions.

Question 30

Question: Which of the following is the commutative law of Boolean algebra?

- a. $A + B = B + A$
- b. $A \cdot B = B \cdot A$
- c. $A + (B \cdot C) = A \cdot B + A \cdot C$
- d. Only (a) and (b)

Correct Answer: Select (d)
Explanation: The commutative laws state that the order of terms does not change the result for addition ($A + B = B + A$) and multiplication ($A \cdot B = B \cdot A$).

Question 31

Question: A system has two components in series, which means the failure of either component will cause the system to fail. The probability of failure for the first component is 10^{-3} and the probability of failure for the second component is 10^{-4}. What is the reliability of this system?

- a. 0.9272
- b. 0.9486
- c. 0.9989
- d. 0.9167

Correct Answer: Select (c)
Explanation: For series reliability:
$R_1 = 1 - 10^{-3} = 0.999$
$R_2 = 1 - 10^{-4} = 0.9999$
$R_{system} = R_1 \times R_2 = 0.999 \times 0.9999 = 0.9989$.

Question 32

Question: The failure curve in reliability engineering is called:

- a. Hazard rate curve
- b. Mean time between failures (MTBF)
- c. Bathtub curve
- d. Hazard curve

Correct Answer: Select (c)
Explanation: The failure rate curve over time for machinery/components typically displays infant mortality, constant failure rate during useful life, and wear-out, forming a bathtub shape.

Question 33

Question: The minimum cut sets for the fault tree below are:

- a. B_4 and $B_3 \cdot B_1$
- b. B_4 and $B_3 + B_1$

- c. E_1 and $E_2 \cdot B_4$
- d. B_1 and $B_3 \cdot B_4 \cdot B_2$

Correct Answer: Select (d)
Explanation: In order to determine the minimum cut sets, we represent the fault tree by a Boolean equation.
Step 1: E_2 and E_1 are connected to the top event (TE) through an OR gate. Therefore, we can write: $TE = E_1 + E_2$.
Step 2: Substitute for E_1 and E_2: $TE = (B_1 \cdot B_2) + (B_1 + E_3)$.
Step 3: $TE = (B_1 \cdot B_2) + B_1 + (B_3 \cdot B_4 \cdot E_4)$.
Step 4: Substitute for E_4: $TE = (B_1 \cdot B_2) + B_1 + (B_3 \cdot B_4 \cdot B_4 \cdot B_2)$.
Step 5: Using idempotent law ($A \cdot A = A$): $B_3 \cdot B_4 \cdot B_4 \cdot B_2 = B_3 \cdot B_4 \cdot B_2$.
Therefore: $TE = (B_1 \cdot B_2) + B_1 + (B_3 \cdot B_4 \cdot B_2)$.
Step 6: Using the law of absorption ($A + A \cdot B = A$): $B_1 \cdot B_2 + B_1 = B_1$.
Therefore: $TE = B_1 + B_3 \cdot B_4 \cdot B_2$.
The two terms B_1 and $B_3 \cdot B_4 \cdot B_2$ are the minimal cut sets.

Question 34

Question: The first step in management of safety within an organization is to:

- a. Appoint a safety and health officer.
- b. Develop a policy statement.
- c. Provide training to workers who may be exposed to hazards on their jobs.
- d. Develop a safety manual.

Correct Answer: Select (b)
Explanation: Development of a policy statement must be the first step in successful management of safety within that organization. This policy statement must clearly define who is responsible, who has the authority, and who will be held accountable.

Question 35

Question: Different styles of management have a direct influence on the flow of information within an organization. In participative management style, the flow of information within an organization is:

- a. From top to bottom
- b. From bottom to top
- c. In all directions
- d. Horizontal

Correct Answer: Select (c)
Explanation: One of the advantages of participative management is that the flow of information within the organization is in all directions. In this style of management, the managers have access to better information as a basis for their decisions.

Question 36

Question: One of the major advantages of an external safety audit is its:

- a. Relatively low cost
- b. Objectivity
- c. Lack of familiarity with the facility
- d. Lack of familiarity with facility personnel

Correct Answer: Select (b)
Explanation: External audits have the clear advantage of objectivity, that is, it provides an opportunity for an independent evaluation. High cost and minimum or no follow-up are among the disadvantages of this type of safety audit.

Question 37

Question: An employee working for a manufacturing facility is responsible for cleaning and maintenance of a storage tank which cannot support life due to its oxygen deficiency. In order to perform these tasks, the employee must enter the tank. Which of the following is a requirement which must be set forth by the management of this organization?

- a. Obtain a permit prior to entry.
- b. Restrict access by unauthorized personnel.
- c. Use proper PPE.
- d. All of the above.

Correct Answer: Select (d)
Explanation: The management must set forth an in-house permitting program for employees to obtain a permit prior to entry into any confined spaces. These areas must be inaccessible to unauthorized personnel and use of proper PPE is required.

Question 38

Question: Organizations that pay workers' compensation premiums recognize such expense as:

- a. Indirect cost of accidents
- b. Uninsured cost of accidents
- c. Hidden cost of accidents
- d. Insured cost of accidents

Correct Answer: Select (d)
Explanation: These costs are definite and known to the organization, and therefore, they comprise the insured cost of the total cost associated with an accident.

Question 39

Question: The fundamental way any organization's management must function is to:

- a. Ensure that the safety and health of all employees are protected.
- b. Ensure compliance with all federal, state, and local occupational safety and environmental health laws, regulations, and standards.
- c. Make decisions and have those decisions implemented.
- d. Ensure that employees have an opportunity to take part in the decision making process.

Correct Answer: Select (c)
Explanation: Management fulfills its responsibilities of managing by making decisions and implementing those decisions. In order to make sound management decisions, the management must have access to facts and information. Management styles greatly affect whether or not a manager has all the right information to make a decision.

Question 40

Question: McGregor has proposed two theories, X and Y, describing an employee's attitude toward his/her work. Theory Y states that:

- a. People are not generally interested in their work, and external motivation is required.
- b. Workers should be motivated by raises and benefit plans.
- c. People are generally interested in their work and are self-motivated.
- d. It is the management's responsibility to motivate employees.

Correct Answer: Select (c)
Explanation: Theory Y states that people are generally interested in their work and are self-motivated. Theory X, on the other hand, states the opposite, i.e., people are not generally interested in their work and are not self-motivated.

Question 41

Question: One of the responsibilities of management in safety as well as other aspects of an organization's business is to follow the 3E's principle. The 3E's are:

- a. Engineering, Education, Enforcement
- b. Effectiveness, Essentials, Efforts
- c. Effectiveness, Education, Enforcement
- d. Establishment, Effectiveness, Education

Correct Answer: Select (a)
Explanation: The 3E's in management stands for Engineering, Education, and Enforcement.

Question 42

Question: Eighteen injuries and illnesses were recorded on OSHA form 300 in one calendar year. Assuming that the company has 100 employees who have worked 50 weeks, 40 hours per week, the OSHA Incidence Rate is:

- a. 18
- b. 12
- c. 24
- d. 6

Correct Answer: Select (a)
Explanation: $IR = \frac{(\text{number of incidences})(200,000)}{\text{actual number of hours worked}} = \frac{(18)(200,000)}{(100)(50)(40)} = 18$.

Question 43

Question: LWDI is an acronym for which of the following?

- a. Lost Wages Due to Injury
- b. Lost Workday and Disability Injury Rate
- c. Labor Work Discipline Incidence
- d. Lost Workday Incidences

Correct Answer: Select (d)
Explanation: LWDI is an acronym for Lost Workday Incidences.

Question 44

Question: Based on principles of Total Quality Management (TQM):

- a. Top management is responsible for product quality.
- b. Line supervisors are responsible for product quality.
- c. Every employee is responsible for product quality.
- d. Only (a) and (b) above.

Correct Answer: Select (c)
Explanation: Based on the principles of TQM, every employee is responsible for product quality and its continual improvement.

Question 45

Question: In management terminology a flat organization refers to one:

- a. With many levels of management.
- b. With few levels of management.
- c. With ineffective management.
- d. Making only a few products.

Correct Answer: Select (b)
Explanation: A flat organization has few levels of management; for example the management structure in most religious or charity organizations is flat. Tall management structure, on the other hand, has many levels of management. The U.S. government management structure is an example of a tall management organization.

Question 46

Question: Argyris, one of the most recognized behavioral scientists, proposes leveling within an organization as a solution to his conflict theory. Which of the following statements best describes the concept of organizational "leveling" as proposed by Argyris?

- a. Line supervisors alone should make all decisions related to safety performance.
- b. The boss does not make all decisions alone.
- c. Top management alone should make all decisions related to safety performance.

- d. One person, usually the safety officer, must make all decisions related to safety and communicate his/her decisions to everyone within the company.

Correct Answer: Select (b)
Explanation: Leveling, according to Argyris, means that everyone should participate either directly or indirectly in the decision making process. In other words, leveling encourages a participative style of management.

Question 47

Question: Herzberg's "Motivation-Hygiene" behavioral theory breaks down the variables affecting performance into hygiene factors and motivation factors. Which of the following is an example of a hygiene factor?

- a. Sense of achievement
- b. Promotion
- c. Recognition
- d. Company policies

Correct Answer: Select (d)
Explanation: Statements a, b, and c are all motivation factors. Company policies, however, is an example of a hygiene factor.

Question 48

Question: Which of the following statements is an example of a motivation factor according to Herzberg's "Motivation-Hygiene" theory?

- a. Relationships with the boss
- b. Money
- c. Responsibility
- d. Working conditions

Correct Answer: Select (c)
Explanation: Statements a, b, and d are examples of hygiene factors. Responsibility is a good example of a motivation factor.

Question 49

Question: The machines in a shop are to be replaced 5 years from now. It is estimated that at that time the cost of new machines will be $3,000,000. At an interest rate of 6 percent compounded annually, what lump sum of money must the machine shop owner deposit now to be able to pay for the cost of new equipment at the end of 5 years?

- a. $1,672,622
- b. $2,241,774
- c. $2,841,675
- d. $1,852,239

Correct Answer: Select (b)
Explanation: The formula for future worth of money is $F = P(1 + i)^n$.
For the problem at hand: $3,000,000 = P(1 + 0.06)^5 \implies P = \frac{3,000,000}{1.338225} = $2,241,774$.

Question 50

Question: The present worth of 2×10^6 five years from now with an interest rate of 8 per cent compounded annually is:

- a. $1,976,323
- b. $1,361,166
- c. $1,723,872
- d. $2,672,523

Correct Answer: Select (b)
Explanation: $P = \frac{F}{(1 + i)^n} = \frac{2,000,000}{(1.08)^5} = \frac{2,000,000}{1.469328} = $1,361,166$.

Question 51

Question: Your department has decided to make major modifications to improve safety. The cost is estimated to be 4 million dollars that the company can borrow from a bank. If the bank charges an interest rate of 12 percent compounded annually and your company would like to pay the principal and interest in 10 years with the first payment after one year, how much the company's annual payments would be:

- a. $707,936
- b. $565,236
- c. $432,175
- d. $653,129

Correct Answer: Select (a)
Explanation: Using uniform series capital recovery factor $A = P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] = 4,000,000 \left[\frac{0.12(1.12)^{10}}{(1.12)^{10} - 1} \right] = $707,936$.

Question 52

Question: If you deposit $2,000 dollars per year in a bank account which pays 7 percent interest compounded annually, how much money do you have after 20 years?

- a. $56,722
- b. $49,723
- c. $81,990
- d. $63,752

Correct Answer: Select (c)
Explanation: Using uniform series future worth factor $F = A \left[\frac{(1+i)^n - 1}{i} \right] = 2,000 \left[\frac{(1.07)^{20} - 1}{0.07} \right] = $81,990$.

Question 53

Question: Which of the following statements BEST describes the doctrine of "Fellow Servant Rule"?

- a. In hazardous waste operations and emergency response, an employee entering a site must always be accompanied by another fellow employee.
- b. The employer is not responsible for injuries suffered by an employee resulting from negligence of another employee.
- c. One of the principles of Total Quality Management which encourage each employee to consider himself/herself personally responsible for providing service to customers.
- d. One of the principles of Total Quality Management which encourages each employee to provide assistance to a fellow employee when possible.

Correct Answer: Select (b)
Explanation: Under the common law Fellow Servant Rule doctrine, an employer could not be held liable for an employee's injury if the injury was caused by the negligence of a co-worker (fellow servant).

Question 54

Question: Which of the following statements BEST describes the role of a safety professional within an organization?

- a. He/she is responsible for the safety conduct of the organization.
- b. He/she is responsible for both occupational and environmental safety conduct of the organization.
- c. He/she develops the information needed for the line to exercise its authority for safety.
- d. He/she is responsible for identifying and eliminating unsafe acts and unsafe conditions within the organization.

Correct Answer: Select (c)
Explanation: Safety professionals serve in a staff advisory role to develop information, tools, and systems that enable line management to exercise authority and accountability for safety.

Question 55

Question: Which of the following best describes a tall organization?

- a. An organization with ineffective management
- b. An organization with many levels of management
- c. An organization with authoritative management
- d. An organization with few levels of management

Correct Answer: Select (b)
Explanation: A tall organization is characterized by many hierarchical levels of management.

Question 56

Question: The actual responsibility for safety within an organization lies with:

- a. Line management
- b. Safety director
- c. Company president/CEO
- d. All employees

Correct Answer: Select (a)
Explanation: The actual responsibility for safety within an organization lies with the line supervisors (first level of management).

Question 57

Question: A linear coefficient of correlation of (-1) between a dependent and an independent variable indicates:

- a. A strong linear correlation
- b. No linear correlation
- c. Data is inaccurate
- d. Variation is all unexplained

Correct Answer: Select (a)
Explanation: The coefficient of linear correlation between two variables lies within ± 1. A coefficient of correlation close to 1 or -1 indicates that the two variables are strongly related. A coefficient of correlation of zero is an indication of no linear relationship.

Question 58

Question: In linear correlation analysis, the coefficient of determination is:

- a. Total variation divided by explained variation
- b. Unexplained variation divided by total variation
- c. Explained variation divided by total variation
- d. Total variation divided by unexplained variation

Correct Answer: Select (c)
Explanation: Coefficient of determination is $r^2 = \frac{\text{explained variation}}{\text{total variation}}$. $\sqrt{r^2} = \pm r$ is called "coefficient of correlation". The coefficient of determination r^2 is a measure of how well the linear relationship equation is performed by two variables.

Question 59

Question: In a safety program study trying to relate lost time injuries to the number of unsafe conditions identified in a given calendar year, the following data has been obtained:
Total variation = 42.95
Explained variation = 21.80
The linear coefficient of correlation is:

- a. ± 0.712
- b. -0.507
- c. $+0.023$
- d. -0.045

Correct Answer: Select (a)
Explanation: Coefficient of determination $r^2 = \frac{21.80}{42.95} = 0.507$.
Coefficient of correlation $r = \sqrt{0.507} = \pm 0.712$.

Note: A value of r close to ± 1 indicates a strong linear relationship, and a value of r close to zero indicates a very weak linear relationship.

Question 60

Question: In a normal probability distribution, large values of standard deviation:

- a. Increase the height of the normal curve and reduce its spread
- b. Decrease the height of the normal curve and reduce its spread
- c. Increase the height of the normal curve and increase its spread
- d. Decrease the height of the normal curve and increase its spread

Correct Answer: Select (d)
Explanation: Large values of standard deviation in normal probability distribution indicate a large spread of the data around the population mean. Therefore, the height of the normal curve is reduced and its spread is increased.

Question 61

Question: The deductible part of an insurance policy is an example of which of the following costs associated with an accident?

- a. Insured cost
- b. Uninsured cost
- c. Direct cost
- d. Premium cost

Correct Answer: Select (b)
Explanation: This is an example of the uninsured cost associated with an accident. The insured (direct) costs of accidents are those that are generally covered by workers' compensation insurance. The uninsured (indirect) costs of accidents are much higher than the direct costs of accidents (by about 2-10 times). Examples of uninsured costs of accidents include, but are not limited to, loss of production, OSHA penalties, property damage, and reduced morale among employees.

Question 62

Question: Which of the following statements is the most correct regarding implementation of a safety and health program?

- a. The cost of preventing accidents must show a return on the profit line or at least avoid expenses.
- b. Accidents must be prevented at any cost.
- c. Accidents must be prevented if they affect the profit more than 10 percent.
- d. Cost of accidents must be absorbed like the cost of any other business function.

Correct Answer: Select (a)
Explanation: In cost-benefit analysis, accident prevention can only be justified if the cost shows a return on profit or at least avoids expenses.

Question 63

Question: The annual accident costs in the U.S. amount approximately to which of the following?

- a. 1 billion dollars
- b. 100 million dollars
- c. 500 million dollars
- d. 100 billion dollars

Correct Answer: Select (d)
Explanation: Occupational accident costs in the U.S. exceed $100 billion annually in direct and indirect expenses.

Question 64

Question: A construction company has a margin of profit of 5 percent on all its contracts. An accident occurs and the company suffers $10,000 in losses. What should the dollar value of jobs that the company bids and is awarded be to recover for this loss?

- a. $50,000
- b. $100,000
- c. $200,000
- d. $500,000

Correct Answer: Select (c)
Explanation: $\text{Required Sales} = \frac{\text{Loss}}{\text{Profit Margin}} = \frac{10,000}{0.05} = $200,000$.

Question 65

Question: A telephone manufacturer suffers an accident loss of $50,000 in its manufacturing facility. If the company sells its telephones at a unit price of $20 with a margin of profit of 10 percent, the number of telephones that must be sold to cover this loss is:

- a. 10,000
- b. 25,000
- c. 50,000
- d. 100,000

Correct Answer: Select (b)
Explanation: $\text{Profit per unit} = 20 \times 0.10 = 2.00.
$\text{Number of units} = \frac{50,000}{2} = 25,000\text{ units}$.

Question 66

Question: In cost-benefit analysis, the dollar value of all costs and benefits associated with a program must be calculated:

- a. On a quantitative basis only
- b. On a qualitative basis only
- c. On a quantitative or qualitative basis

- d. Based on cost indices for each program alternative

Correct Answer: Select (c)
Explanation: Cost-benefit evaluations incorporate both tangible quantitative expenses/savings and intangible qualitative factors.

Question 67

Question: A company must sell $1,000,000 worth of its product to recover $100,000 cost of loss from a recent accident in its manufacturing facility. The company's margin of profit is:

- a. 10%
- b. 20%
- c. 5%
- d. 15%

Correct Answer: Select (a)
Explanation: $\text{Profit Margin} = \frac{\text{Loss}}{\text{Sales Needed}} = \frac{100,000}{1,000,000} = 0.10 = 10\%$.

Question 68

Question: A television manufacturer is trying to recover from $100,000 loss. The company sells its televisions for a unit price of $300. If the company's margin of profit is 5 percent, how many television sets must the company sell to recover from this loss?

- a. 2,000
- b. 1,500
- c. 3,000
- d. 6,666

Correct Answer: Select (d)
Explanation: $\text{Profit per TV} = 300 \times 0.05 = \$15$.
$\text{TVs needed} = \frac{100,000}{15} = 6,666.67 \approx 6,666$.

Question 69

Question: Overtime work required as a result of an accident is an example of which of the following costs associated with an accident?

- a. Direct
- b. Uninsured
- c. Insured
- d. Hidden

Correct Answer: Select (b)
Explanation: Overtime premiums paid to catch up on lost production represent uninsured (indirect) accident costs.

Question 70

Question: The first step in responding to spill of a hazardous substance is to:

- a. Contain the spilled substance
- b. Clean up the spilled substance
- c. Identify the spilled substance
- d. Ensure that people are protected

Correct Answer: Select (d)
Explanation: Life safety and protecting personnel from exposure is always the top priority before containment or cleanup.

Question 71

Question: The evaluation of hazards posed by a spilled substance must be conducted from a safe distance located:

- a. Upwind of the spill
- b. Down wind of the spill
- c. Cross wind of the spill
- d. Only (a) and (c) above

Correct Answer: Select (a)
Explanation: Approaching hazardous chemical releases from upwind prevents toxic vapor or flammable plume exposure.

Question 72

Question: If a spill that exceeds EPA reportable quantities occurs inside a building and is not released to the environment, you must contact:

- a. The National Response Center
- b. The EPA
- c. A person within the organization who is identified in SPCC
- d. The EPA and OSHA

Correct Answer: Select (c)
Explanation: Spills contained entirely inside a facility without environmental release trigger internal SPCC/facility response plan notifications rather than National Response Center reporting.

Question 73

Question: The Shewhart Cycle is a concept which is used extensively in Total Quality Management. Which of the following best describes this cycle?

- a. Concept, act, plan, manage
- b. Plan, do, study, act
- c. Manage, do, act, plan
- d. Do, act, study, manage

Correct Answer: Select (b)
Explanation: The Shewhart (or Deming) Cycle is Plan-Do-Study-Act (PDSA) or Plan-Do-Check-Act (PDCA).

Question 74

Question: Which of the following statements best describes the concept of scientific management?

- a. The job of management is to make decisions and have those decisions implemented.
- b. Management makes its decisions based on the principle of assumption of risk.
- c. Human performance can be defined and controlled through work standards and rules.
- d. Only (a) and (b) above.

Correct Answer: Select (c)
Explanation: Frederick Taylor's Scientific Management holds that tasks can be optimized scientifically and human work controlled through strict work standards and standardized rules.

Question 75

Question: Which of the following best describes Heinrich's theory of accident causation?

- a. Unsafe act, unsafe condition, ineffective management, injury
- b. Injury, technocracy, ineffective management, unsafe act, unsafe condition
- c. Social environment, ineffective management, accident, injury
- d. Social environment, ancestry, unsafe act/condition, accident, injury

Correct Answer: Select (d)
Explanation: Heinrich's Domino Theory identifies five sequential factors: 1. Ancestry/Social Environment, 2. Fault of Person, 3. Unsafe Act or Mechanical/Physical Hazard, 4. Accident, 5. Injury.

Question 76

Question: Which of the following best describes the chronological order of steps involved in devising a Safety By Objective (SBO) program?

- a. Set objective, provide training, monitor the program
- b. Analyze climate, evaluate appraisal system, determine allowable objectives, train, set objectives
- c. Set objectives, provide feedback, provide training, execute, enforce
- d. Determine allowable objective, set objectives, provide training, execute, monitor, enforce

Correct Answer: Select (d)
Explanation: SBO sequence moves from determining allowable organizational objectives to setting targets, training, execution, monitoring, and enforcement.

Question 77

Question: How much money would you have in your account after 8 years if you deposit $1,000 per year starting one year from now at an interest rate of 10% compounded annually?

- a. $9,762
- b. $10,675
- c. $11,435
- d. $14,372

Correct Answer: Select (c)
Explanation: $F = A \left[\frac{(1+i)^n - 1}{i} \right] = 1,000 \left[\frac{(1.10)^8 - 1}{0.10} \right] = 1,000(11.4358) = $11,435$.

Question 78

Question: Your company plans to purchase new equipment for a manufacturing plant 5 years from now. The cost of new equipment is $2,000,000. How much should the company deposit in a bank account one year from now to be able to pay cash for the cost of the new equipment? Assume a fixed interest rate of 8 percent.

- a. $500,168
- b. $405,100
- c. $392,155
- d. $340,912

Correct Answer: Select (d)
Explanation: Using uniform series sinking fund factor: $A = 2,000,000 \left[\frac{0.08}{(1.08)^5 - 1} \right] = $340,912$.

Question 79

Question: How much money should you be willing to pay now for a note that will yield $1,000 per year for 10 years starting next year at an interest rate of 10 percent?

- a. $6,144
- b. $5,293
- c. $7,895
- d. $4,238

Correct Answer: Select (a)
Explanation: $P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] = 1,000 \left[\frac{(1.10)^{10} - 1}{0.10(1.10)^{10}} \right] = $6,144$.

Question 80

Question: If you deposit $5,000 dollars in an IRA account now paying an interest of 10 percent per year, how much money would you have after 15 years?

- a. $11,732
- b. $8,963
- c. $20,886
- d. $9,687

Correct Answer: Select (c)
Explanation: $F = P(1 + i)^n = 5,000(1.10)^{15} = 5,000(4.1772) = $20,886$.

Question 81

Question: You are trying to purchase a new computer system. The salesclerk makes you an offer to either pay $7,000 now or $10,000 five years from now. If the interest rate is 10 percent compounded annually would you pay the $7,000 now?

- a. Yes
- b. No

Correct Answer: Select (b)
Explanation: Present worth of $10,000 in 5 years at 10%: $P = \frac{10,000}{(1.10)^5} = $6,209$. Since $6,209 is less than $7,000, you should NOT pay $7,000 now (answer is No).

Question 82

Question: In safety management theory, accidents in the work place are considered to be:

- a. Caused by failure of man or machine to perform as designed in a given environment.
- b. Symptoms of a management failure.
- c. Caused as a result of physical, chemical, biological, or ergonomics hazards.
- d. All of the above.

Correct Answer: Select (b)
Explanation: In safety management, accidents have both direct and indirect causes. Although it is true that the direct cause of a given accident may be a combination of factors ranging from not controlling hazards to human failure, the basic cause of all accidents goes back to management failure.

Question 83

Question: The primary objective of a workplace safety inspection is to:

- a. Discover situations that are in violation of federal, state, or local laws.
- b. Discover areas that should be included in the employee safety training program.
- c. Identify the probability and severity of hazards.
- d. Identify and correct unsafe conditions and unsafe acts.

Correct Answer: Select (d)
Explanation: According to National Safety Council, 98 percent of all workplace accidents are caused by either unsafe acts or unsafe conditions; the remaining 2 percent are attributed to natural causes such as earthquakes or flooding. The primary goal of any safety inspection program must be to identify and eliminate unsafe acts and unsafe conditions.

Question 84

Question: The OSHA Incidence Rate (IR) for a company with 300 employees and 9 recordable work related injuries and illnesses and 600,000 work hours is:

- a. 5
- b. 3
- c. 4
- d. 2

Correct Answer: Select (b)
Explanation: $IR = \frac{(\text{injuries/illnesses})(200,000)}{\text{actual hours worked}} = \frac{9 \times 200,000}{600,000} = 3$.

Question 85

Question: According to Heinrich's estimates, 88 per cent of all occupational accidents are caused as a result of:

- a. Unsafe conditions
- b. Natural disasters
- c. Unsafe acts
- d. Only (a) and (c) above

Correct Answer: Select (c)
Explanation: Heinrich reviewed more than 75,000 accident cases and concluded that about 88 percent of accident cases reviewed were caused by unsafe acts of people.

Question 86

Question: A standard which indicates the end result to be achieved regardless of the procedures used in called:

- a. Voluntary standard
- b. Specification standard
- c. Performance standard
- d. Mandatory standard

Correct Answer: Select (c)
Explanation: Performance standards only specify the end result that needs to be achieved. Many employers prefer this type of standard because of the freedom that they have in meeting the standard. Specification standards, on the other hand, specify the detailed requirements such as dimensions.

Question 87

Question: Organizational safety must begin with development and implementation of a policy. Which of the following is NOT part of an organization's safety policy statement?

- a. Authority
- b. Hazard analysis
- c. Responsibility
- d. Accountability

Correct Answer: Select (b)
Explanation: Safety policy statements must clearly define the organization's position in regard to safety issues. It must assign responsibility, authority, and accountability for safety related matters. Hazard analysis is an engineering/operational procedure, not part of policy statements.

Question 88

Question: Most organizations have two major components; line and staff. Which of the following is an example of the staff component?

- a. Manufacturing shift foreman
- b. Security personnel
- c. Sales personnel
- d. Manufacturing workers

Correct Answer: Select (b)
Explanation: Line elements of an organization are those who are involved in producing a product or delivering a service while the staff elements take care of other business matters such as legal, security, finance, etc.

Question 89

Question: Staff and line members within an organization perform different functions. Line managers are involved with producing a product or delivering a service to clients while staff members assist in special matters such as finance, legal, and security. Safety must be part of:

- a. Line element of an organization only
- b. Line supervisors only
- c. Both line and staff functions
- d. Staff element only

Correct Answer: Select (c)
Explanation: The organization safety policy statement must clearly express that safety is part of everyone's job (both line and staff).

Question 90

Question: Which of the following statements best describes the concept of functional authority?

- a. Line supervisors are responsible for safety matters in their department.
- b. The ultimate responsible party for safety related matters within a company are the President and Chief Executive Officer (CEO) of the company.
- c. The Vice-President for manufacturing writes a letter to the safety director. The letter states that the safety director can stop any operation that he feels poses an imminent danger to the safety and health of employees.
- d. The company safety policy statement must clearly assign accountability, responsibility, and authority in regard to the functional safety performance of the organization.

Correct Answer: Select (c)
Explanation: Functional authority grants a staff specialist (such as a safety director) direct authority over specific functional matters (such as stopping imminent danger operations) outside their normal staff advisory role.

Question 91

Question: The PRIMARY objective of any accident investigation must be to:

- a. Prevent future similar accidents
- b. Identify the causes of accident
- c. Collect evidence for legal purposes
- d. Determine the equipment malfunction(s) and or human error(s) that caused the accident

Correct Answer: Select (a)
Explanation: The primary objective is to prevent occurrence of similar accidents in future. Determining the cause(s) of the accident is a first step towards accomplishing this objective.

Question 92

Question: The two major types of accident investigation are:

- a. Fault Tree and HAZOP analysis
- b. Primary and secondary
- c. Pre-accident and post-accident events
- d. General and special

Correct Answer: Select (d)
Explanation: General investigations use generic procedures and usually do not require special techniques or instruments. Special investigations, on the other hand, require special knowledge such as fire investigation or aviation accident investigations.

Question 93

Question: An organization that the management closely follows the chain of command is an example of a/an:

- a. Authoritative structure
- b. Participative structure
- c. Inefficient structure
- d. Formal structure

Correct Answer: Select (d)
Explanation: This is an example of a formal management structure. A formal structure can be defined as an organization that is hierarchical in nature. The formal structure of a company can usually be found on its organizational chart. This chart normally identifies each staff member with his or her job title.

Question 94

Question: Which of the following is NOT an element of 3-E's Principle?

- a. Education
- b. Efficiency
- c. Engineering
- d. Enforcement

Correct Answer: Select (b)
Explanation: The elements of 3-E's Principals are engineering, education, and enforcement.

Question 95

Question: "Just-in-time" inventory is used for which of the following purposes?

- a. Deliver products to customers in a timely manner.
- b. Ensure that raw materials are available for production.
- c. Reduce quantities of stored hazardous materials.
- d. Prepare a list of hazardous chemicals for compliance with OSHA's Hazard Communication Standard.

Correct Answer: Select (c)
Explanation: "Just-in-time" inventory manages delivery of hazardous substances when they are needed on site, and therefore, minimizes unnecessary storage of hazardous substances.

Question 96

Question: It is the legal responsibility of management to ensure that employee exposure to airborne hazardous substances is kept below:

- a. Threshold Limit Values (TLV)
- b. Permissible Exposure Limits (PEL)
- c. Short Term Exposure Limits (STEL)
- d. Threshold Limit Value Ceiling (TLVC)

Correct Answer: Select (b)
Explanation: PELs are set by OSHA, and compliance with established PELs for different chemicals is mandatory. OSHA sets these enforceable PELs to protect workers against the health effects of exposure to hazardous substances. PELs are regulatory limits on the amount or concentration of a substance in the air.

OSHA PELs are based on an eight-hour time-weighted average (TWA).

Question 97

Question: The MAJOR objective of a safety inspection is to:

- a. Identify and correct unsafe conditions and unsafe acts
- b. Ensure compliance with federal, state, and local regulations
- c. Measure the effectiveness of the safety program
- d. Identify any PPE which is needed in each work area

Correct Answer: Select (a)
Explanation: A safety inspection must focus on identifying and correcting unsafe conditions and unsafe acts. According to National Safety Council statistics, 98% of industrial accidents are caused by either unsafe conditions or unsafe acts. 2% of industrial accidents are caused as a result of natural disasters such as an earthquake.

Question 98

Question: The cost of an injury or illness is divided into insured costs and uninsured costs. The total cost of an injury or illness is determined by:

- a. Only considering the uninsured costs.
- b. Subtracting the insured costs from the uninsured costs.
- c. Adding the insured costs to the uninsured costs.
- d. Subtracting the insured cost from the uninsured costs and dividing the result by the insured costs.

Correct Answer: Select (c)
Explanation: $\text{Total Cost} = \text{Insured Costs} + \text{Uninsured Costs}$.

Question 99

Question: In workers' compensation terminology, the no fault system means:

- a. Regardless of who is at fault the worker will be paid for workplace injuries and illnesses.
- b. The employer is protected against lawsuits.
- c. The worker can only be paid upon proof of negligence on the part of the employer.
- d. Only a and b above.

Correct Answer: Select (d)
Explanation: No-fault workers' compensation guarantees benefits to injured workers regardless of fault, while granting employers immunity from common-law negligence tort lawsuits.

Question 100

Question: An employee has been modifying the safety and health data after they have been approved by the company safety professional holding a CSP. Which approach is the most appropriate to take?

- a. Verbally fire the employee on the spot.
- b. Write a letter to company headquarters requesting permission to investigate the case.
- c. Call OSHA.
- d. Set up a meeting with the employee and the management in order to investigate the case.

Correct Answer: Select (d)
Explanation: Proper management due process requires investigating the situation factually in a meeting with the employee and management before taking formal disciplinary actions.

Question 101

Question: Which of the following statements BEST describes an OSHA Performance Standard?

- a. Employers with incidence rates of 2.5 or greater must upgrade their safety performance or are in violation of this standard.
- b. Specific objectives must be obtained regardless of a specific method.
- c. Specific objectives must be obtained using OSHA approved methods.
- d. Only (a) and (c) above.

Correct Answer: Select (b)
Explanation: Performance standards mandate the required objective or result to be achieved, leaving the specific compliance methods to the employer's discretion.

Question 102

Question: A Type I error in the testing of a hypothesis indicates:

- a. A valid hypothesis was rejected.
- b. An invalid hypothesis was accepted.
- c. Standard deviation is zero.
- d. Variance is zero.

Correct Answer: Select (a)
Explanation: In statistical hypothesis testing, a Type I error occurs when a true (valid) null hypothesis is incorrectly rejected.

Question 103

Question: The Managerial Grid, developed by Blake and Mouton, evaluates managerial leadership style on two axes: concern for production and concern for people. A 9,1 leadership style represents:

- a. Country Club Management
- b. Task or Authoritarian Management
- c. Team Management
- d. Impoverished Management

Correct Answer: Select (b)
Explanation: On Blake and Mouton's Managerial Grid, 9,1 represents high concern for production (9) and low concern for people (1), which is Task or Authoritarian management.

Question 104

Question: On the Blake and Mouton Managerial Grid, a 1,9 leadership style represents:

- a. Country Club Management
- b. Task Management
- c. Team Management
- d. Impoverished Management

Correct Answer: Select (a)
Explanation: A 1,9 rating indicates low concern for production (1) and high concern for people (9), known as Country Club management.

Question 105

Question: On the Blake and Mouton Managerial Grid, a 9,9 leadership style represents:

- a. Country Club Management
- b. Task Management
- c. Team Management
- d. Impoverished Management

Correct Answer: Select (c)
Explanation: A 9,9 rating reflects maximum concern for both production and people, known as Team management.

Question 106

Question: Which of the following statements BEST describes the Otis Test?

- a. It is used in quality control to distinguish between patent defect and latent defect in a product batch.
- b. It is a graph of budgeting vs. scheduling that measures a manager's effectiveness.
- c. It is used to measure attitudes, preferences, and subjective reactions.
- d. It is a test that evaluates abstract thinking and a person's ability to reason (IQ test).

Correct Answer: Select (d)
Explanation: The Otis Test is used to measure an individual's IQ. This is a test that evaluates abstract thinking and a person's ability to reason. It is a multiple choice test that focuses on measuring quantitative, spatial, and verbal reasoning ability.

Question 107

Question: Cohesiveness in group dynamics is best defined as:

- a. The establishing of rules and policies necessary for group operation.
- b. The degree that members of a team or group desire to remain in the team or group.
- c. The ability of a team leader to direct the group.
- d. The level of productivity achieved by a group.

Correct Answer: Select (b)
Explanation: Cohesiveness is defined as the degree that members of a team or group desire to remain in the team or group.

Question 108

Question: Which of the following statements BEST describes the Likert Scale?

- a. It is used in quality control to distinguish between patent defect and latent defect in a product batch.
- b. It is a graph of budgeting vs. scheduling that measures a manager's effectiveness.
- c. It is used to measure attitudes, preferences, and subjective reactions.
- d. It is a site map about accessibility, groupware, and usability.

Correct Answer: Select (c)
Explanation: A Likert Scale is used to measure attitudes, preferences, and subjective reactions. For example, in software design, it can often subjectively measure efficiency and effectiveness with performance metrics such as time taken or errors made.

Question 109

Question: In the Hawthorne Studies, investigators found no strong relationship between workplace conditions and worker productivity. However, the investigators reached several conclusions. Which of the following is NOT a conclusion of the Hawthorne Studies?

- a. Sentiments and participation can reduce resistance to change.
- b. The workplace is an interlocking social system, not simply a production system.
- c. Employees should not be considered appendages of machinery.
- d. The Hawthorne Studies opened the door to the study of client-centered therapy, small group behavior, organizational theory, and research methodology.

Correct Answer: Select (d)
Explanation: Statement (d) describes general historical impacts, whereas the specific core conclusions of the Hawthorne Studies focused on human social factors in industrial workplaces (statements a, b, and c).

Question 110

Question: A company with 5 lost workdays (LWD) in a calendar year and 200 employees working 50 weeks in a calendar year for 40 hours a week has an LWDI of:

- a. 6.3
- b. 2.5
- c. 4.0
- d. 7.5

Correct Answer: Select (b)
Explanation: $\text{LWDI} = \frac{(\text{LWD})(200,000)}{\text{total hours worked}} = \frac{5 \times 200,000}{200 \times 50 \times 40} = \frac{1,000,000}{400,000} = 2.5$.

Question 111

Question: A manufacturing company had an LWDI equal to 4 in a given calendar year. The company has 200 employees who, on the average, worked 40 hours per week for 50 weeks. What is the actual number of lost workdays for this company?

- a. 3
- b. 4
- c. 5

- d. 8

Correct Answer: Select (d)
Explanation: $\text{LWDI} = \frac{(\text{LWD})(200,000)}{\text{total hours}} \implies 4 = \frac{(\text{LWD})(200,000)}{200 \times 40 \times 50} \implies 4 = \frac{(\text{LWD})(200,000)}{400,000} \implies \text{LWD} = 8$.

Question 112

Question: The LWDI for a company in a given calendar year was equal to 5. Each employee worked for 50 weeks for an average of 40 hours per week. Assuming that the actual number of lost day cases was 10, the number of employees working for this company is:

- a. 200
- b. 250
- c. 300
- d. 400

Correct Answer: Select (a)
Explanation: $\text{LWDI} = \frac{(\text{cases})(200,000)}{\text{hours}} \implies 5 = \frac{10 \times 200,000}{N \times 50 \times 40} = \frac{2,000,000}{2,000 N} \implies 5 = \frac{1,000}{N} \implies N = 200$.

Question 113

Question: Application of water-based foam to a flammable liquid fire has the potential for creating which of the following?

- a. Boiling liquid expanding vapor explosion (BLEVE)
- b. High-volume mixture of steam, air, and burning fuel
- c. Phosgene gas due to breakdown of halon 1301 in water-based foam
- d. Large amount of toxic carbon dioxide gas

Correct Answer: Select (b)
Explanation: Assuming that the surface temperature of the burning liquid is above the boiling point of water, this application has the potential for creating a very high volume of an air, fuel, and steam mixture above the surface of the burning liquid.

Question 114

Question: A cab company conducted a study of their accident frequency for 200 days. At the conclusion of the study, it was found that the probability of NOT having an accident is 0.99. This means that, on the average, the cab company experiences an accident every _____ days.

- a. 20
- b. 100
- c. 200
- d. 50

Correct Answer: Select (b)
Explanation: The probability of having an accident is $1 - 0.99 = 0.01$ or $\frac{1}{100}$, which equals one accident every 100 days.

Question 115

Question: Management commitment, site analysis, hazard prevention and control, and safety and health training are the four major components of which of the following programs?

- a. OSHA Voluntary Protection Program
- b. OSHA Hazard Communication Program
- c. EPA Toxic Substances Control Program
- d. OSHA Emergency Response Program

Correct Answer: Select (a)
Explanation: This program is an optional program and has the four components named in the question.

The VPP recognize employers and workers in the private industry and federal agencies who have implemented effective safety and health management systems and maintain injury and illness rates below national Bureau of Labor Statistics averages for their respective industries. In VPP, management, labor, and OSHA work cooperatively and proactively to prevent fatalities, injuries, and illnesses through a system focused on: hazard prevention and control; worksite analysis; training; and management commitment and worker involvement.

Question 116

Question: When setting safety and environmental goals, the goals must have all of the following qualities EXCEPT:

- a. Be measurable
- b. Be reportable
- c. Be achievable
- d. Have a time frame

Correct Answer: Select (b)
Explanation: The goals and objectives must be realistic, specific, measurable, achievable, and time-bound for the program at hand.

Question 117

Question: A coefficient of correlation of 0.95 between two variables indicates that:

- a. The variables are almost directly proportional to each other.
- b. The variables are almost inversely proportional to each other.
- c. There is no correlation between the two variables.
- d. No statistical inferences can be made about the variables under study.

Correct Answer: Select (a)
Explanation: The coefficient of correlation varies between -1 and +1. A coefficient of correlation of -1 indicates a perfect inverse relationship between the variables. A coefficient of correlation of +1 indicates a perfect direct

relationship between the two variables. In this case, with a coefficient of correlation of 0.95, we can conclude that the two variables have an almost perfect direct relationship between them.

Question 118

Question: A company has five plants at different geographical locations. At the conclusion of the same comprehensive safety training program given at each plant, the same test was given to trainees to measure their level of understanding of the materials covered in the training program. Subsequently, the same outside consulting firm to measure the safety performance at each facility conducted a two-month audit at each facility. The audit results were ranked on a scale of 1 to 10, with 1 being the worst and 10 being the best safety performance. The results are tabulated below:

Plant	Average Test Score	Safety Performance
1	72	7
2	67	7
3	81	9
4	82	8
5	72	5

Based on the data presented, the coefficient of correlation is:

- a. 0.275
- b. -0.275
- c. 0.658
- d. 0.346

Correct Answer: Select (c)
Explanation: Calculating Pearson's r for the 5 data pairs yields $r = 0.658$.
Discussion: The coefficient of correlation varies between -1.0 and +1.0. A coefficient of correlation equal to -1.0 indicates perfect inverse relationship between variables under study and +1.0 indicates a perfect direct relationship. In this case, we can conclude that the safety training had a positive impact on safety performance but that there is room for improvement.

Question 119

Question: The coefficient of correlation for two variables is 0.95 at 0.01 significance level. This means that:

- a. The two variables have a strong direct relationship but there is a 1% chance that the conclusion is wrong.
- b. The two variables have a strong inverse relationship but there is a 99% chance that the conclusion is wrong.
- c. The two variables have a strong inverse relationship but there is a 1% chance that the conclusion is wrong.
- d. The two variables have a strong direct relationship but there is a 99% chance that the conclusion is wrong.

Correct Answer: Select (a)
Explanation: $r = 0.95$ indicates a strong direct positive relationship, and a 0.01 significance level ($
lpha = 0.01$) means there is a 1% probability of type I error (falsely rejecting a true null hypothesis).

Question 120

Question: Which of the following statements is true?

- a. OHSAS 18000 is an international occupational health and safety management system specification.
- b. OHSAS 18000 was created via a concerted effort from a number of the world's leading national standards bodies, certification bodies, and specialist consultancies.
- c. OHSAS 18000 actually forms part of the acclaimed Health and Safety Electronic Kit.
- d. All of the above.

Correct Answer: Select (d)
Explanation: All statements are true in regard to OHSAS 18000. Also, companies that use such management systems are subject to audits.

Question 121

Question: Which of the following is considered to be the best approach in an audit process?

- a. When possible, auditors should verify in person that management processes are implemented.
- b. Verify compliance by interviewing the management team.
- c. Verify compliance by interviewing employees.
- d. Verify compliance by interviewing both management and employees.

Correct Answer: Select (a)
Explanation: Direct physical verification and field observation of operational compliance is superior to relying solely on interview testimony.

Question 122

Question: Which of the following can best achieve impartiality during an audit process?

- a. Not having any line or operational responsibility for the task
- b. Understanding the code of ethics for the organization
- c. Setting the goals and objectives for the audit process
- d. Having complete management support

Correct Answer: Select (a)
Explanation: Ensuring auditors have no direct operational, line, or administrative responsibility for the areas being audited guarantees objective, independent evaluation.

Question 123

Question: The following data was compiled after a one-year study at ABC plant. What is the expected cost of all incidents in a given year?

Probability	Incident Cost
0.15	$1,000
0.20	$5,000

0.35	$7,000
0.30	$2,000

- a. $4,200
- b. $7,500
- c. $3,700
- d. $6,800

Correct Answer: Select (a)
Explanation: $\text{Expected Cost} = (0.15 \times 1,000) + (0.20 \times 5,000) + (0.35 \times 7,000) + (0.30 \times 2,000) = 150 + 1,000 + 2,450 + 600 = $4,200$.

Question 124

Question: Which of the following is required to be independent of line management?

- a. Safety training personnel
- b. Safety audit teams
- c. Contractors
- d. All of the above

Correct Answer: Select (b)
Explanation: Safety audit teams must maintain independence from line management to ensure objective auditing without conflicts of interest.

Question 125

Question: All of the following statements are true in regards to Pareto charts and the Pareto principle EXCEPT:

- a. The probability of risk can be calculated from the area under the curve.
- b. 20% of the problems cause 80% of the risk.
- c. A Pareto chart can be constructed by segmenting the range of data into groups.
- d. A Pareto chart is used to graphically summarize and display the relative importance of the differences between groups of data.

Correct Answer: Select (a)
Explanation: Calculating risk probabilities from area under the curve applies to continuous probability density functions (like normal distributions), not discrete bar-graph Pareto distributions.

Question 126

Question: Which of the following describes management leadership and effective employee participation as crucial factors contributing to the success of safety and health programs within organizations?

- a. ANSI Z10
- b. EPA Form R
- c. Institute of Safety and Systems Management
- d. Center for Occupational Safety and Health

Correct Answer: Select (a)
Explanation: ANSI Z10 (Occupational Health and Safety Management Systems) highlights management leadership and employee participation as foundational pillars.

Question 127

Question: Which of the following ISO standards deals with managing and conducting quality environmental audits?

- a. ISO 2001
- b. ISO 17008
- c. ISO 19011
- d. ISO 18007

Correct Answer: Select (c)
Explanation: ISO 19011 provides guidelines for auditing management systems, including quality and environmental management systems.

Question 128

Question: When selecting jobs for Job Safety Analysis, all of the following are important factors to consider EXCEPT:

- a. Jobs that have high probability of leading to an incident
- b. Jobs that have potential for causing a high severity incident
- c. New jobs
- d. Jobs that are stressful to some employees

Correct Answer: Select (d)
Explanation: Jobs should not be selected randomly or solely due to individual stress. Priority for JSA goes to accident frequency, accident severity potential, newly established jobs, or modified processes.

Question 129

Question: The root cause of any workplace incident/accident is considered to be:

- a. A number of interacting factors
- b. Lack of sufficient employee training
- c. Lack of proper communication
- d. Poor management

Correct Answer: Select (d)
Explanation: Workplace accidents are considered to be symptoms of poor management (management system failures).

Question 130

Question: A project is to be carried out in three phases. The error rate for Phase One is 0.34%, for Phase Two is 0.54%, and for Phase Three is 0.24%. The mean error rate for this project is:

- a. 0.0037
- b. 0.0560
- c. 0.0024
- d. 0.0067

Correct Answer: Select (a)
Explanation: The mean error rate is obtained by converting percentages to decimals and averaging: $\text{Mean} = \frac{0.0034 + 0.0054 + 0.0024}{3} = \frac{0.0112}{3} = 0.003733 \approx 0.0037$.

Question 131

Question: Which of the following is the most important consideration in developing a data collection system?

- a. Ensure that the data is compatible for statistical use.
- b. Ensure that statistical inferences can be drawn from the data.
- c. Define how the data is going to be used.
- d. Ensure that the data represent the population under study.

Correct Answer: Select (c)
Explanation: Defining the subsequent use of data is the most important consideration in design of a data collection system.

In the design of a data collection system, consultation with data users and providers as well as choice of data requirements must be considered:

- What data items are needed and how they will be used
- The precision level required for estimates
- The format, level of detail, and types of tabulations
- When and how frequently users need the data

Question 132

Question: All of the following are elements of an effective loss control system EXCEPT:

- a. Hazard identification
- b. Assessment of risks associated with the hazards
- c. Establishment of chain of custody procedures
- d. Development of means to control the risks

Correct Answer: Select (c)
Explanation: Loss control systems that do not address hazard identification, risk assessment, and risk control means are incomplete and ineffective. Chain of custody is a legal/forensic procedure, not a core loss control element.

Question 133

Question: Which of the following best describes the most important step in an incident investigation program?

- a. Collect meaningful data for subsequent analysis.
- b. Preserve evidence and interview witnesses.
- c. Ensure that appropriate regulatory agencies are notified.
- d. Determine the casual factors and root causes that led to the incident.

Correct Answer: Select (d)
Explanation: One of the major goals of an effective incident investigation program must be to prevent the occurrence of similar incidents. This can be accomplished by determining the casual factors and their subsequent root causes that led to the incident.

Question 134

Question: Which of the following is NOT an element of an effective system safety program plan?

- a. Define the elements and their needs for the mission.
- b. Identify means to improve the safety performance of an existing system.
- c. Identify safety deficiencies.
- d. Determine the Mean Time Between Failures (MTBF) for the system.

Correct Answer: Select (d)
Explanation: The System Safety Program Plan (SSPP) is a critical document for subsequent system safety analyses. It must address management tasks, system safety engineering tasks, deficiency identification, and improvement methods. Calculating specific MTBF values is a reliability engineering task rather than an SSPP structural element.

Question 135

Question: Which of the following is used to demonstrate the difference between actual expenditure at a given time and the projected budget for a project?

- a. Gantt chart
- b. Budget variance report
- c. Budget slack report
- d. Budget bar chart

Correct Answer: Select (b)
Explanation: A budget variance report compares actual expenditures against planned project budgets to highlight overruns or underruns.

Question 136

Question: In a manufacturing facility, the management has established a goal to have no lost time injury for 500 days. This is an example of:

- a. Benchmarking
- b. Safety coaching
- c. Mission statement

- d. All of the above

Correct Answer: Select (a)
Explanation: Setting specific quantitative safety performance targets against standard metrics is an example of safety benchmarking.

Question 137

Question: Which of the following statements in regard to incidence rates is INCORRECT?

- a. Incidence rates are excellent indicators of an organization's safety performance.
- b. Incidence rates are lagging indicators of safety performance.
- c. Overall, incidence rates alone cannot measure safety performance.
- d. The standard for calculating incidence rates is an organization with 100 employees who work 40 hours per week for 50 weeks a year.

Correct Answer: Select (a)
Explanation: Statement (a) is incorrect because incidence rates are lagging indicators reflecting past failures, and relying on them alone provides an incomplete measure of safety performance.

Question 138

Question: Which of the following BEST describes the purpose of ISO 9000?

- a. The ISO 9000 family is primarily concerned with quality management.
- b. The ISO 9000 family is primarily concerned with environmental management.
- c. The ISO 9000 family is primarily concerned with proper management of occupational safety within organizations.
- d. The ISO 9000 family is primarily concerned with the customer's quality requirements.

Correct Answer: Select (a)
Explanation: The ISO 9000 family standards focus specifically on quality management systems.

Question 139

Question: Which of the following statements BEST describes the overall purpose of ISO 14000?

- a. The ISO 14000 family is primarily concerned with quality management.
- b. The ISO 14000 family is primarily concerned with environmental management.
- c. The ISO 14000 family is primarily concerned with proper management of occupational safety within organizations.
- d. The ISO 14000 family is primarily concerned with the customer's quality requirements.

Correct Answer: Select (b)
Explanation: The ISO 14000 family addresses environmental management systems.

Question 140

Question: The organizational chart shown in company documentation displaying CEO -> Vice Presidents -> Managers -> Employees in direct descending boxes is an example of a:

- a. Matrix chart
- b. Functional chart
- c. Hierarchical chart
- d. Operational chart

Correct Answer: Select (c)
Explanation: A traditional scalar chart descending from CEO down through managers to staff is a hierarchical organizational chart.

Question 141

Question: All of the following statements are true in regard to Pareto charts EXCEPT:

- a. A Pareto chart is a type of chart that contains both bars and a line graph.
- b. Individual values are represented in descending order by bars.
- c. The cumulative total is represented by the line.
- d. The left vertical axis shows the cumulative percentage of the total number of occurrences.

Correct Answer: Select (d)
Explanation: On a Pareto chart, the left vertical axis represents frequency of occurrence or cost, while the right vertical axis represents cumulative percentage.

Question 142

Question: According to OSHA, which of the following injuries or illnesses must be recorded on the OSHA 300 log?

- a. Injuries that require attention beyond first aid
- b. All first aid injuries
- c. Minor scratches requiring a bandage
- d. Temporary headaches relieved by non-prescription aspirin

Correct Answer: Select (a)
Explanation: According to OSHA regulations, only work-related injuries and illnesses requiring medical treatment beyond first aid are recordable.

Question 143

Question: Based on the information presented in Table 1 below, and assuming that each employee works 8 hours per day, 5 days a week, 50 weeks in a calendar year, the LWDI for Plant C in 2004 is:

Table 1: Injury Information for Company X in 2004

Plant	# Injuries Treated with First Aid	# Injuries Treated Beyond First Aid	# Lost Work Days Due to Injuries	# Employees
A	25	10	80	400
B	35	15	120	600
C	40	30	200	900

- a. 6.66
- b. 22.22
- c. 4.50
- d. 18.32

Correct Answer: Select (b)
Explanation: $\text{LWDI} = \frac{\text{lost work days} \times 200,000}{\text{total hours}} = \frac{200 \times 200,000}{900 \times 8 \times 5 \times 50} = \frac{40,000,000}{1,800,000} = 22.22$.

Question 144

Question: Based on the information presented in Table 1 above, and assuming that each employee works 8 hours per day, 5 days a week, 50 weeks in a calendar year, the OSHA Incident Rate (IR) for recordable injuries in 2004 in Plant A is:

- a. 8.50
- b. 2.50
- c. 5.00
- d. 8.00

Correct Answer: Select (b)
Explanation: Recordable injuries for Plant A = 10 (treated beyond first aid).
$\text{Hours} = 400 \times 2,000 = 800,000$.
$IR = \frac{10 \times 200,000}{800,000} = 2.50$.

Question 145

Question: Based on the information presented in Table 1, the total number of recordable injuries for all plants in 2004 is:

- a. 155
- b. 85
- c. 55
- d. 400

Correct Answer: Select (c)
Explanation: Recordable injuries are those treated beyond first aid: $10 + 15 + 30 = 55$.

Question 146

Question: Which of the following statements best describes the process of adsorption?

- a. Removal of a component of a liquid mixture by another liquid

- b. Removal of a component of a gas mixture by a liquid
- c. Removal of a component of a liquid mixture by a gas
- d. Removal of one or more components of a gas mixture on a solid bed

Correct Answer: Select (d)
Explanation: Adsorption is the retention of gas/vapor molecules on the surface of a solid sorbent bed (such as activated carbon).

Question 147

Question: Factors that affect safety effectiveness include behavioral influence as well as motivational factors. Which of the following is an example of behavioral influence?

- a. Achievement
- b. Advancement
- c. Church
- d. Responsibility

Correct Answer: Select (c)
Explanation: Social institutions such as church, family, and peer groups exert external behavioral influence on worker attitudes and actions.

Question 148

Question: An employee is recognized by his boss for excellent safety performance. This is an example of an:

- a. Intrinsic reward
- b. Extrinsic reward
- c. Internal reward
- d. Only (a) and (c) above

Correct Answer: Select (b)
Explanation: Extrinsic rewards originate externally from supervisors, management, or peers (such as praise, bonuses, or awards).

Question 149

Question: Which of the following statements is an example of a motivation factor according to Herzberg's "Motivation-Hygiene" theory?

- a. Raise in salary
- b. Responsibility
- c. Company policy
- d. More vacation time

Correct Answer: Select (b)
Explanation: Responsibility is a intrinsic motivator under Herzberg's framework, whereas salary, policy, and vacation are hygiene factors.

Question 150

Question: Being independent of line management is a requirement for which of the following?

- a. Safety training personnel
- b. Safety audit teams
- c. Contractors
- d. All of the above

Correct Answer: Select (b)
Explanation: Safety audit teams must operate independently of line supervision to maintain impartiality.

Question 151

Question: All of the following statements are true in regard to Pareto charts and the Pareto principle EXCEPT:

- a. The probability of risk can be calculated from the area under the curve
- b. 20% of the problems cause 80% of the risk
- c. A Pareto chart can be constructed by segmenting the range of data into groups
- d. A Pareto chart is used to graphically summarize and display the relative importance of the differences between groups of data

Correct Answer: Select (a)
Explanation: Risk probabilities are not derived from area under the curve on a Pareto chart.

Question 152

Question: Which of the following statements is correct in regard to OHSAS 18001?

- a. Calls for continuous improvement in safety and health management systems
- b. Sets standards to predict the financial health of an organization
- c. Sets standards for proper environmental management and in particular proper disposal of hazardous wastes
- d. Sets standards for pollution prevention

Correct Answer: Select (a)
Explanation: OHSAS 18001 specifies requirements for an occupational health and safety management system emphasizing continual improvement.

Question 153

Question: The safety office of Company XYZ reports an average of four lost time injuries in the past three months. What is the probability that there will be three or less lost time injuries in the next three months?

- a. 0.2754
- b. 0.3265

- c. 0.4332
- d. 0.14618

Correct Answer: Select (c)
Explanation: Using Poisson distribution with mean $\lambda = 4$:
$P(X \le 3) = P(0) + P(1) + P(2) + P(3)$
$P(0) = \frac{e^{-4} 4^0}{0!} = 0.0183$
$P(1) = \frac{e^{-4} 4^1}{1!} = 0.0733$
$P(2) = \frac{e^{-4} 4^2}{2!} = 0.1465$
$P(3) = \frac{e^{-4} 4^3}{3!} = 0.1954$
$P(X \le 3) = 0.0183 + 0.0733 + 0.1465 + 0.1954 = 0.4335 \approx 0.4332$.

Question 154

Question: According to the Harvard Business Review, one of the reasons people resist change is excess uncertainty. Which of the following statements is NOT an example of excess uncertainty?

- a. A change that feels like walking off a cliff blindfolded
- b. A change that feels like heading toward an unknown
- c. A change that interferes with autonomy and can make people feel that they have lost control over their territory
- d. A change that can be described by the saying "Better the devil you know than the devil you don't know."

Correct Answer: Select (c)
Explanation: Statement (c) describes loss of control / autonomy, which is a distinct reason for resisting change separate from excess uncertainty.

Question 155

Question: The balanced scorecard developed by Robert Kaplan and David Norton of Harvard Business School suggests that we view organizations from four perspectives. Which of the following is NOT one of the four perspectives used in scorecards?

- a. Financial perspective
- b. Customer perspective
- c. Internal business process perspective
- d. Technical perspective

Correct Answer: Select (d)
Explanation: The four Balanced Scorecard perspectives are: Financial, Customer, Internal Business Processes, and Learning & Growth. Technical perspective is not one of them.

Question 156

Question: During one-on-one manager-employee progress reviews, reviewing obstacles encountered since the last review and identifying ways to handle them occurs in which phase of the performance review?

- a. Preparation phase
- b. Discussion/coaching phase

- c. Closing phase
- d. Evaluation phase

Correct Answer: Select (b)
Explanation: Reviewing performance obstacles and coaching solutions occurs during the interactive discussion/coaching phase.

Question 157

Question: Regulatory, Warning, and Information are the three basic categories of workplace signs. The subcategory "Danger" falls under:

- a. Regulatory
- b. Warning
- c. Information
- d. Caution

Correct Answer: Select (b)
Explanation: Danger and Caution subcategories both fall under Warning signs (indicating immediate/definite or potential hazards).

Question 158

Question: In the Shewhart (PDCA) cycle, documenting changes, informing others, and making recommendations occurs in which phase?

- a. Planning
- b. Doing
- c. Checking
- d. Acting

Correct Answer: Select (d)
Explanation: In the Acting phase, results are documented, stakeholders informed, and process recommendations adopted.

Question 159

Question: A hospital patient wears a wrist band barcoded with medication data. A nurse scans the wristband and medication barcode using a mobile computer. If the wrong medication or dosage is selected, an alarm sounds. This is an example of:

- a. Poka-yoke (error proofing)
- b. Statistical process control
- c. Six Sigma
- d. JIT inventory

Correct Answer: Select (b)
Explanation: Barcode scanning verification systems act as automated point-of-use administrative controls preventing medication errors.

Question 160

Question: When initiating a product recall, actions taken AFTER publicly announcing the recall include:

- a. Assembling recall committee
- b. Quarantining inventory
- c. Identifying affected lots
- d. Monitoring recall effectiveness and processing returned products

Correct Answer: Select (d)
Explanation: Public announcement marks the formal recall launch; subsequent steps involve tracking returns and evaluating recall success.

Question 161

Question: The five levels of Maslow's hierarchy of needs in ascending order are:

- a. Safety, physiological, social, esteem, self-actualization
- b. Physiological, safety, belonging/love, esteem, self-actualization
- c. Physiological, social, safety, esteem, self-actualization
- d. Safety, social, esteem, physiological, self-actualization

Correct Answer: Select (b)
Explanation: Maslow's hierarchy ascends: 1. Physiological, 2. Safety, 3. Belonging/Love, 4. Esteem, 5. Self-actualization.

Question 162

Question: Job Safety Analysis (JSA) is usually completed on a form containing three columns headed:

- a. Task, Hazard, Solution
- b. Job Step, Potential Injury, Control
- c. Job Step, Hazard, Control
- d. Task, Risk, PPE

Correct Answer: Select (d)
Explanation: Standard JSA forms utilize three columns: Job Step, Hazard, and Controls.

Question 163

Question: In probability theory, if drawing a black ball from a bag containing one black, one white, and one green ball after having already drawn the green ball, this is an example of:

- a. Conditional probability
- b. A priori probability
- c. Subjective probability

- d. Probability distribution

Correct Answer: Select (a)
Explanation: The outcome probability depends on the prior event (drawing the green ball), defining conditional probability.

Question 164

Question: According to OSHA, the ratio of indirect to direct costs of accidents varies from a high of 20:1 to a low of:

- a. 10:1
- b. 5:1
- c. 1:1
- d. 2:1

Correct Answer: Select (d)
Explanation: According to OSHA estimates, indirect-to-direct cost ratios range from 20:1 down to 1:1 depending on severity.

Question 165

Question: Heinrich's domino theory states that an accident is one factor in a sequence leading to injury. The factor IMMEDIATELY preceding the accident in the domino sequence is:

- a. Ancestry/social environment
- b. Fault of person
- c. Unsafe act or condition
- d. Injury

Correct Answer: Select (d)
Explanation: In Heinrich's 5-domino sequence (Ancestry -> Fault of Person -> Unsafe Act/Condition -> Accident -> Injury), the unsafe act or condition immediately precedes the accident.

Question 166

Question: In an organizational structure where employees report to a functional department manager as well as a project manager, this chart represents a:

- a. Matrix organization
- b. Hierarchical organization
- c. Flat organization
- d. Line organization

Correct Answer: Select (a)
Explanation: Matrix management assigns personnel to dual reporting channels (functional department and project team).

Question 167

Question: The Hawthorne studies suggested that worker productivity gains occurred due to the impact of _____ effects on the workers.

- a. Emotional
- b. Motivational
- c. Behavioral
- d. Environmental

Correct Answer: Select (b)
Explanation: Productivity increases stemmed from motivational effects created by management interest and attention paid to workers.

Question 168

Question: Which of the following is NOT part of the process of Management By Objectives (MBO)?

- a. Objectives are defined jointly by management and employees
- b. The organization must monitor performance measures
- c. The plan can be modified when needed
- d. It is a tightly controlled top-down system where workers have no input

Correct Answer: Select (d)
Explanation: MBO relies on joint goal setting and employee participation, not rigid autocratic top-down control.

Question 169

Question: All of the following statements are true regarding Program Evaluation and Review Technique (PERT) EXCEPT:

- a. It is a statistical tool
- b. It analyzes the tasks involved in completing a project
- c. It is not suitable for planning and scheduling large and complex projects
- d. It specially focuses on the time needed to complete each task in a project

Correct Answer: Select (c)
Explanation: PERT was developed specifically to simplify planning and scheduling for large, complex projects.

Question 170

Question: All of the following are examples of OSHA horizontal standards EXCEPT:

- a. Fire protection
- b. The construction industry
- c. First aid
- d. Working surfaces

Correct Answer: Select (b)
Explanation: Horizontal standards apply across all industries (fire protection, first aid). Construction (29 CFR 1926) is a vertical standard applying to a specific industry.

Question 171

Question: All of the following are part of safety in the design of playgrounds EXCEPT:

- a. Playgrounds must be designed for one age group.
- b. Equipment with moving parts must be separated from the rest of the playground.
- c. For high surfaces, guard rails and protective barriers must be installed.
- d. Children must be under adult supervision while in the playground.

Correct Answer: Select (a)
Explanation: Playgrounds are designed for different age groups with separate designated zones for younger vs older children.

Question 172

Question: According to the Consumer Products Safety Commission (CPSC), which of the following statements is true regarding a public playground safety checklist?

- a. Make sure surfaces around playground equipment have at least 12 inches of wood chips, mulch, sand, or pea gravel, or mats made of safety-tested rubber or rubber-like materials.
- b. Check that protective surfacing extends at least 6 feet in all directions from play equipment. For swings, be sure surfacing extends, in back and front, twice the height of the suspending bar.
- c. Make sure play structures more than 30 inches high are spread at least 9 feet apart.
- d. All of the above

Correct Answer: Select (d)
Explanation: All listed specifications are standard CPSC public playground safety guidelines.

Question 173

Question: Which of the following IS NOT considered an essential function of a management team?

- a. Dependency on each other
- b. Open communication
- c. Having a real reason for existing
- d. Provide required safety training to subordinates

Correct Answer: Select (d)
Explanation: Providing direct routine safety training is a line supervisor task, not an essential executive management team design function.

Question 174

Question: Which of the following has the direct responsibility and accountability to ensure that the company's health and safety policies are properly enforced?

- a. Safety director
- b. Line management
- c. Top management
- d. All of the above

Correct Answer: Select (b)
Explanation: Line management holds direct responsibility and operational accountability for enforcing safety policies on the shop floor.

Question 175

Question: Which of the following IS NOT an essential part of an effective employee performance appraisal interview?

- a. Indicating that all employees may have problems and that this is not unusual.
- b. Helping the employee to take a positive approach to his/her problems.
- c. Ensuring that the performance review is based on factors that can be measured objectively.
- d. Identifying the nature of the problems and why they happened.

Correct Answer: Select (d)
Explanation: Performance reviews focus on objective evaluation and constructive development rather than punitive forensic fault-finding.

Question 176

Question: The _____ of a project is defined as the work that, when properly completed, results in the success of the project.

- a. Critical path
- b. Scope
- c. Sound management
- d. Contingency

Correct Answer: Select (b)
Explanation: Project scope defines the total work required to successfully complete the project deliverables.

Question 177

Question: In project management, a hierarchical decomposition of work with specific objectives and deliverables at each level is called:

- a. Work Breakdown Structure
- b. Work Decomposition Plan
- c. Formal Decomposition Plan
- d. Goal Identification Plan

Correct Answer: Select (a)
Explanation: A Work Breakdown Structure (WBS) decomposes project scope into manageable, deliverable-oriented components.

Question 178

Question: Gantt charts used in project scheduling are also called:

- a. Pie charts
- b. Matrix charts
- c. Bar charts
- d. Time charts

Correct Answer: Select (c)
Explanation: Gantt charts display project schedule tasks as horizontal bar charts over a time scale.

Question 179

Question: When significant operational or process changes are considered, the management should place priority on which of the following?

- a. Perform a change analysis to understand the impacts of changes being considered.
- b. Perform a fault tree analysis to understand the possible incident scenarios.
- c. Ensure that all employees are informed about the impacts of changes on safety.
- d. Provide training for employees who are affected by implementing the changes.

Correct Answer: Select (a)
Explanation: Performing a formal Management of Change (MOC) hazard analysis prior to implementation is the top safety priority.

Question 180

Question: Which of the following statements is NOT true?

- a. Line supervisors are interested in doing those tasks that they think are being measured.
- b. Line supervisors are more interested in performing those tasks that they think they will be rewarded for doing.
- c. Line supervision should be provided with written organization policies for enforcement and accountability.
- d. Line supervisors are equally interested in performing all of their duties whether they are measured or not.

Correct Answer: Select (d)
Explanation: Behavioral management principles demonstrate that supervisors prioritize tasks that are measured and rewarded over unmeasured tasks.

Question 181

Question: Which of the following statements BEST describes the role or function of a safety committee?

- a. The committee must have meetings at least once a week.
- b. The committee must have at least one representative from each department within the organization.
- c. The committee must set goals and objectives and measure its performance against achieving those goals and objectives.
- d. The committee must have at least one member who is familiar with safety features of all processes used in the organization.

Correct Answer: Select (c)
Explanation: An effective safety committee must establish realistic, measurable goals and monitor its progress toward achieving them.

Question 182

Question: Benchmarking can be a very effective tool in safety training. Which of the following statements best describes the process of benchmarking?

- a. Using information about other companies' training practices.
- b. Identifying potential training needs.
- c. Identifying what the trainees should do after the training process.
- d. Determining how the company's business strategy influences safety training.

Correct Answer: Select (a)
Explanation: Benchmarking involves comparing organizational practices and performance metrics against industry best practices and leading peer companies.

Question 183

Question: All of the following are elements of a successful communication process EXCEPT:

- a. Sender
- b. Message
- c. Interpretation
- d. Receiver

Correct Answer: Select (c)
Explanation: The classic communication loop consists of Sender, Message, Channel, Receiver, and Feedback. Interpretation is an individual cognitive process, not a structural loop element.

Question 184

Question: ISO 14001 deals with which of the following?

- a. Sets out the criteria for an environmental management system
- b. Sets out a framework for safety management in the workplace
- c. Sets out standards for minimizing of global warming
- d. All of the above

Correct Answer: Select (a)
Explanation: ISO 14001 specifies core criteria for establishing an environmental management system.

Question 185

Question: Safety and Health is the official journal of which of the following organizations?

- a. American Society of Safety Engineers (ASSE)
- b. American Industrial Hygiene Association (AIHA)
- c. National Safety Council (NSC)
- d. Board of Certified Safety Professionals (BCSP)

Correct Answer: Select (c)
Explanation: Safety + Health magazine is published monthly by the National Safety Council. (Note: ASSE publishes Professional Safety and AIHA publishes Synergist).

Question 186

Question: NEISS is an acronym for:

- a. National Electronic Injury Surveillance System
- b. National Emergency Information Support System
- c. Navigational Emergency Information Support System
- d. National Employee Information Support System

Correct Answer: Select (a)
Explanation: NEISS stands for National Electronic Injury Surveillance System, operated by the U.S. Consumer Product Safety Commission (CPSC).

Question 187

Question: The major objective(s) of injury data collection within an organization is:

- a. To make comparisons of the progress of the safety program
- b. To analyze methods of approach to safety
- c. To assist in settlement of Workers' Compensation cases
- d. All of the above.

Correct Answer: Select (d)
Explanation: Injury data collection serves to monitor safety progress, evaluate hazard control approaches, and support workers' compensation claims management.

Question 188

Question: The run chart is one of the most powerful and simple statistical tools for data analysis. The run chart is a:

- a. Plot of data over a period of time
- b. Plot of the mean as a function of the standard deviation
- c. Plot of experimental runs as a function of experimental results

- d. Plot of equipment failures as a function of operating conditions

Correct Answer: Select (a)
Explanation: A run chart is a line graph plotting process observations/data sequentially over time.

Question 189

Question: Which of the following techniques is more commonly used to analyze accident investigation data?

- a. HAZOP
- b. Preliminary Hazard Analysis
- c. What If Analysis
- d. MORT

Correct Answer: Select (d)
Explanation: Management Oversight and Risk Tree (MORT) is a structured analytical logic tree designed specifically for in-depth accident investigations.

Question 190

Question: Use of OSHA incident rate in evaluation of the effectiveness of a safety and health program is an example of:

- a. Qualitative measurement
- b. Legal measurement
- c. Cause-consequence measurement
- d. Quantitative measurement

Correct Answer: Select (d)
Explanation: Incidence rates provide numerical, quantitative safety performance measurements.

Question 191

Question: Which of the following is NOT a necessary element of a valid contract?

- a. Amount of investment by each party.
- b. Legal matters associated with the contract.
- c. Definition of mutuality of obligations.
- d. Subject matter of the contract.

Correct Answer: Select (a)
Explanation: Essential contract elements are offer/acceptance, mutual obligation, consideration, legal capacity, and lawful subject matter. Specific investment amounts are not required.

Question 192

Question: The employer must record information, on OSHA Form 300, about every work-related injury or illness that involves ANY of the following EXCEPT:

- a. Loss of consciousness
- b. Job transfer
- c. Restricted work activity
- d. Treatment by first aid only

Correct Answer: Select (d)
Explanation: OSHA explicitly exempts work-related injuries or illnesses treated solely by first aid from recordkeeping on Form 300.

Question 193

Question: In order to create a Total Safety Culture within an organization, the management must pay attention to three major areas. Which of the following is NOT one of these three major areas?

- a. Behavioral factors
- b. Engineering factors
- c. Environmental factors
- d. Personal factors

Correct Answer: Select (b)
Explanation: Total Safety Culture frameworks focus on Person, Behavior, and Environment factors. Engineering is considered an environmental/system element.

Question 194

Question: There are many unique characteristics about a real team which lead to success. Which of the following IS NOT true regarding a real team?

- a. Members share common purpose and goals
- b. Members are often evaluated as individuals
- c. Members hold themselves mutually accountable
- d. Members process complimentary skills

Correct Answer: Select (b)
Explanation: Real teams hold themselves mutually accountable and are evaluated on collective team outputs, whereas workgroups evaluate members individually.

Question 195

Question: Which of the following is NOT considered to be a component of team performance?

- a. Attitude
- b. Behavior
- c. Reinforcement
- d. Knowledge

Correct Answer: Select (c)
Explanation: Team training and performance components consist of Knowledge, Attitude, and Behavior (KAB). Reinforcement is a management feedback mechanism.

Question 196

Question: The coefficient of correlation is often used to determine the nature of relationship between two variables such as the relationship between safety inspections and the number of unsafe conditions or unsafe acts. The coefficient of correlation is a range of variance between:

- a. Zero (0) and plus one (+1)
- b. Negative one (-1) and zero (0)
- c. Negative one (-1) and plus one (+1)
- d. Plus one (+1) and plus two (+2)

Correct Answer: Select (c)
Explanation: Correlation coefficients range strictly from -1.0 to +1.0.

Question 197

Question: The number of safety inspections and the number of on the job injuries have a coefficient of correlation of -0.95 in company XYZ. Which of the following statements is true?

- a. Safety inspections can help to reduce the number of injuries.
- b. Safety inspections actually cause the number of injuries to increase.
- c. Safety inspections have no significant effect on the number of injuries.
- d. The coefficient of correlation can not be negative. It is a number between 0 and +1.

Correct Answer: Select (a)
Explanation: $r = -0.95$ indicates a strong inverse relationship: as safety inspections increase, workplace injuries significantly decrease.

Question 198

Question: *The Synergist* is a monthly periodical that provides information on occupational and environmental health. Which of the following organizations publishes this periodical?

- a. The American Industrial Hygiene Association
- b. The American Conference of Governmental Industrial Hygienists
- c. The Environmental Protection Agency
- d. The National Safety Council

Correct Answer: Select (a)
Explanation: *The Synergist* is published monthly by the American Industrial Hygiene Association (AIHA).

Question 199

Question: *Professional Safety* is the official publication of the:

- a. System Safety Society
- b. American Society of Safety Engineers
- c. American Society of Professional Safety Engineers
- d. National Safety Council

Correct Answer: Select (b)
Explanation: *Professional Safety* is the official peer-reviewed journal of the American Society of Safety Engineers (ASSE, now ASSP).

Question 200

Question: Nowadays, many employers support telecommuting. This can offer cost benefits to the company and time and lifestyle benefits to the employee. By far, the most common location for telecommuting is:

- a. Hotels
- b. Satellite offices of the company
- c. Employee's car
- d. Employee's home

Correct Answer: Select (d)
Explanation: The employee's home is by far the primary telecommuting work location (>80%).

Question 201

Question: All of the following are examples of spreadsheet software EXCEPT:

- a. Excel
- b. VisiCalc
- c. PowerPoint
- d. Lotus

Correct Answer: Select (c)
Explanation: Microsoft PowerPoint is a presentation graphic software application, not a spreadsheet program.

Question 202

Question: All of the following are BCSP professional membership organizations EXCEPT:

- a. Society of Fire Protection Engineers
- b. Institute of Industrial Engineers
- c. American Board of Industrial Hygiene
- d. System Safety Society

Correct Answer: Select (c)
Explanation: ABIH (now BOSH) is a certifying board for industrial hygienists, not a individual professional membership society.

Question 203

Question: Which of the following provides the most efficient way to make Material Safety Data Sheets available to employees within an organization?

- a. Internet
- b. Microsoft Internet Explorer
- c. HTTP
- d. LAN

Correct Answer: Select (d)
Explanation: Local Area Networks (LAN) provide reliable, rapid, internal enterprise-wide electronic access to SDS documents.

Question 204

Question: Which of the following organizations is responsible for certifying industrial hygiene laboratories?

- a. ASSE
- b. ABIH
- c. AIHA
- d. ABHL

Correct Answer: Select (c)
Explanation: The American Industrial Hygiene Association (AIHA) administers laboratory accreditation programs for industrial hygiene testing.

Question 205

Question: Which of the following statements is true in regard to the Poisson distribution?

- a. 65 percent of the population falls within four standard deviations.
- b. It is used to model the number of events occurring within a given time interval.
- c. It requires a sample size of at least 100.
- d. It is a continuous probability distribution.

Correct Answer: Select (b)
Explanation: The Poisson distribution is a discrete probability distribution used to model the number of independent events occurring within a specified time or space interval.

Question 206

Question: In system safety, MORT stands for:

- a. Management Oversight and Risk Tree
- b. Management Operation and Risk Tree

- c. Management Oversight and Risk Technology
- d. Management Operation and Risk Technology

Correct Answer: Select (a)
Explanation: MORT stands for Management Oversight and Risk Tree.

Question 207

Question: The primary purpose of a safety policy statement is to:

- a. Inform employees that safety is important
- b. Set up safety committees
- c. Define safety responsibility, authority, and accountability
- d. Satisfy OSHA requirements

Correct Answer: Select (c)
Explanation: A safety policy statement establishes management commitment and explicitly defines safety responsibility, authority, and accountability.

Question 208

Question: The ANSI Z10 standard covers:

- a. Personal protective equipment
- b. Occupational Health and Safety Management Systems
- c. Confined space entry
- d. Eye and face protection

Correct Answer: Select (b)
Explanation: ANSI/AIHA Z10 defines requirements for occupational health and safety management systems.

Question 209

Question: In project management, the critical path is:

- a. The shortest path through a project network
- b. The path with the most slack
- c. The longest sequence of activities that determines the minimum project duration
- d. The path containing non-critical activities

Correct Answer: Select (c)
Explanation: The critical path is the longest path of dependent activities through a project network, dictating the total project completion time.

Question 210

Question: In managing many projects, a chart is often used that identifies the responsible role, accountable role, consulted role, and informed role for each task of the project. This chart is called:

- a. RACI chart
- b. Project matrix chart
- c. ICRA chart
- d. ARCI chart

Correct Answer: Select (a)
Explanation: RACI stands for Responsible, Accountable, Consulted, and Informed matrix.

Question 211

Question: In project management, comparison of actual performance with potential or desired performance is called:

- a. Goal analysis
- b. Performance evaluation
- c. Gap analysis
- d. Slack analysis

Correct Answer: Select (c)
Explanation: Gap analysis involves comparing actual current performance against desired potential or ideal performance.

Question 212

Question: A simple accident causation model groups the causes of any accident into five categories. Which of the following are elements of these categories?

- a. Material
- b. Management
- c. Personnel
- d. All of the above

Correct Answer: Select (d)
Explanation: The 5M accident causation model categories are: Task (Media), Material, Machinery, Personnel (Man), and Management.

Question 213

Question: Which of the following IS NOT an example of types of group health plans in the workplace?

- a. Fully-insured group health insurance plans
- b. Self-insured health insurance plans
- c. Self-insured medical reimbursement plans
- d. Partially-insured group health plans

Correct Answer: Select (d)
Explanation: Health plans are structured legally as fully-insured or self-insured (self-funded) plans; "partially-insured group health plans" is not a recognized formal classification.

Question 214

Question: Management theories are implemented to optimize productivity and service quality of an organization. All of the following statements in regards to management theories are correct EXCEPT:

- a. In "Contingency theory" managers make decisions based on the situation at hand.
- b. In "Systems theory" managers recognize how different systems affect a worker and how a worker affects the systems around them.
- c. In "Chaos theory" change is considered to be inevitable and is rarely controlled.
- d. Theory "y" assumes that workers are not naturally driven and do not want to take responsibility.

Correct Answer: Select (d)
Explanation: Statement (d) describes Theory X. Theory Y assumes workers are naturally motivated and seek responsibility.

Question 215

Question: Accident models have evolved over time and can be characterized by three models. All of the following are considered to be accident models EXCEPT:

- a. Sequence of events model
- b. MORT
- c. Epidemiological or latent failure model
- d. Systemic model

Correct Answer: Select (b)
Explanation: MORT is an accident investigation tool/technique, not one of the three historical accident model classes (sequential, epidemiological, systemic).

Question 216

Question: Which of the following statements is/are true in regards to ANSI Z10 standard?

- a. The Z10 standard defines minimum requirements for an occupational safety and health management system.
- b. The Z10 standard sets forth standards for proper selection and use of hard hats.
- c. The Z10 standard covers new steps for lockout/tagout procedures.
- d. The Z10 standard sets forth requirements for a sound environmental management system.

Correct Answer: Select (a)
Explanation: ANSI/AIHA Z10 provides an overall blueprint and minimum requirements for occupational safety and health management systems.

Question 217

Question: Which of the following statements is/are true in regards to OHSAS 18000?

- a. It sets forth an environmental management system to optimize management efficiency.
- b. It is a family of international standards relating to occupational health and safety management.
- c. It sets forth new standards for sound management techniques in construction industry.
- d. All of the above.

Correct Answer: Select (b)
Explanation: OHSAS 18000 is an international occupational health and safety management system specification.

Question 218

Question: Which of the following statements is/are true in regards to ISO 19011?

- a. It is an international standard for proper management of accident investigations.
- b. It is an international standard that sets forth guidelines for management systems auditing.
- c. It is an environmental management system for proper management of hazardous wastes operations.
- d. All of the above.

Correct Answer: Select (b)
Explanation: ISO 19011 sets forth guidelines for management systems auditing (quality and environmental).

Question 219

Question: Under the "management of change" element of the Process Safety Management standard, the employer is responsible for implementing written procedures to manage changes. These procedures must ensure that all of the following issues are addressed EXCEPT:

- a. The technical basis for the proposed change.
- b. Modifications to operating procedures.
- c. Necessary time period.
- d. The effect of change on the compliance audits program.

Correct Answer: Select (d)
Explanation: PSM Management of Change (MOC) addresses technical basis, safety/health impacts, operating procedure modifications, and authorization timeframes. Compliance audit schedules are governed separately under 1910.119(o).

Question 220

Question: Which of the following has "plan, do, check, and act" as four of its basic elements?

- a. OHSAS 20001
- b. OHSAS 18001
- c. OHSAS 19001
- d. OHSAS 17001

Correct Answer: Select (b)
Explanation: OHSAS 18001 is structured around the Plan-Do-Check-Act (PDCA) continual improvement cycle.

Question 221

Question: Which of the following statements is true about confined spaces?

- a. Always have toxic chemicals in their atmosphere
- b. Have limited means for entry or exit
- c. They must not be entered without an air purifying respirator.
- d. All the above

Correct Answer: Select (b)
Explanation: A confined space is defined as having limited or restricted means for entry or exit and not being designed for continuous employee occupancy.

Question 222

Question: OSHA standards require safe means of access and egress to all excavations. Ladders, steps, ramps, or other safe means of exit must be provided for workers working in trench excavations 4 feet (1.22 meters) or deeper. These devices must be located within _____ feet of all workers.

- a. 10
- b. 25
- c. 50
- d. 5

Correct Answer: Select (b)
Explanation: OSHA 29 CFR 1926 Subpart P requires safe means of egress in trenches 4 feet or deeper requiring no more than 25 feet of lateral travel for workers.

Question 223

Question: Which of the following is NOT one of the four protective measures to ensure excavation safety?

- a. Soil stabilization
- b. Benching
- c. Sloping
- d. Shoring

Correct Answer: Select (a)
Explanation: The primary protective system methods specified under OSHA Subpart P are Benching, Sloping, Shoring, and Shielding (soil stabilization is not classified as one of the four primary systems).

Question 224

Question: OSHA requires the use of fall protection when construction workers are working at heights of _____ feet or greater above a lower level.

- a. 4
- b. 10
- c. 6
- d. 8

Correct Answer: Select (c)
Explanation: In construction (29 CFR 1926.501), the fall protection threshold trigger is 6 feet above a lower level (general industry trigger is 4 feet).

Question 225

Question: Which of the following statements is correct regarding safe use of ladders?

- a. Always maintain a 2-point (one hand and a foot) contact on the ladder when climbing.
- b. An extension or straight ladder used to access an elevated surface must extend at least 3 feet above the point of support.
- c. The proper angle for setting up a ladder is to place its base a third of the working length of the ladder.
- d. All the above.

Correct Answer: Select (b)
Explanation: Extension/straight ladders accessing upper landing surfaces must extend at least 3 feet (0.9 m) above the support point. (Climbing requires 3-point contact; base distance rule is 1/4 working length).

Question 226

Question: A ladder that is 20 feet from the base to the top support must be placed in such a way that the base distance from the vertical plane of the top support is:

- a. 10 ft.
- b. 5 ft.
- c. 2 ft.
- d. 1 ft.

Correct Answer: Select (b)
Explanation: The 4:1 ladder angle rule requires base distance to be 1/4 of working length: $\frac{20}{4} = 5\text{ feet}$.

Question 227

Question: Fall protection systems can be categorized as "passive" or "active". There are two types of active fall protection systems, fall arrest and _____ systems:

- a. Stationary
- b. Netting
- c. Point fall protection
- d. Fall restraint

Correct Answer: Select (d)
Explanation: Active fall protection systems require worker participation and harness/lanyard equipment, divided into Fall Arrest and Fall Restraint systems.

Question 228

Question: An employee working for a manufacturing facility is responsible for cleaning and maintenance of a storage tank which cannot support life due to its oxygen deficiency. To perform these tasks, the employee must enter the tank. Which of the following is a requirement which must be set forth by the management of this organization?

- a. Obtain a permit prior to entry.
- b. Restrict access by unauthorized personnel.
- c. Use proper PPE.
- d. All the above.

Correct Answer: Select (d)
Explanation: Management must establish an in-house permit-required confined space program mandating entry permits, access restriction, and appropriate PPE.

Question 229

Question: Which of the following must be the first step to be taken prior to entry into a confined space?

- a. Ensure that the appropriate type of Personal Protective Equipment is available.
- b. Conduct a lockout/tagout.
- c. Test the atmosphere.
- d. Notify the emergency personnel.

Correct Answer: Select (c)
Explanation: Before entry into a permit space, testing the internal atmosphere for oxygen content, flammable gases/vapors, and potential toxic air contaminants is the mandatory first step.

Question 230

Question: A "Lock-out/Tag-out" system is used to:

- a. Keep people from stealing your tools
- b. Prevent accidental contact with electrical current
- c. Keep you from completing your work
- d. Add another step to your work

Correct Answer: Select (b)
Explanation: Lockout/tagout procedures isolate hazardous energy sources (electrical, mechanical, hydraulic) to prevent accidental energization or shock during service/maintenance.

Question 231

Question: OSHA requires [1910.212(a)(1)] that the employer to provide one or more methods of machine guarding to protect the operator and other employees from hazardous mechanical motion or action. Which of the following IS NOT an example of hazardous mechanical motion or action requiring machine guarding?

- a. Point of operation
- b. Ingoing nip points
- c. Rotating parts
- d. Vibrating parts

Correct Answer: Select (d)
Explanation: Machine guarding standards address mechanical pinch points, rotating shafts, and points of operation. Vibration hazards are addressed by dampening or vibration isolation.

Question 232

Question: _____ machine guards shut off or disengage power and prevent the machine from starting when the guard is opened.

- a. Fixed
- b. Adjustable
- c. Interlocked
- d. Self-adjusting

Correct Answer: Select (c)
Explanation: Interlocked barrier guards automatically cut machine power or disengage drive mechanisms when opened.

Question 233

Question: OSHA regulations require that the Confined Space Entry Plan for Permit-Required Situations be reviewed, and if necessary, updated:

- a. Every 2 years.
- b. Annually.
- c. After an emergency.
- d. Annually and after an emergency.

Correct Answer: Select (d)
Explanation: Permit-required confined space entry programs must be reviewed annually and immediately following any emergency or program deficiency.

Question 234

Question: OSHA requires that the minimum distance between a crane and any overhead obstruction be:

- a. 1 foot
- b. 5 feet
- c. 12 inches
- d. 3 inches

Correct Answer: Select (a)
Explanation: OSHA standards require maintaining a minimum clearance of at least 1 foot between cranes/booms and overhead structural obstructions.

Question 235

Question: Which of the following is the main purpose of lockout/tagout procedures?

- a. To prevent unauthorized entry into secured areas.
- b. To ensure that electrical equipment perform as their design intent.
- c. To prevent voltage surcharges which might damage the equipment.
- d. To isolate equipment or machinery that use electricity from their power sources before maintenance or servicing work.

Correct Answer: Select (d)
Explanation: The fundamental purpose of LOTO is isolating machinery/equipment from energy sources prior to servicing or maintenance.

Question 236

Question: All the following are components of a personal fall arrest system EXCEPT:

- a. Anchor
- b. Sling
- c. Body harness
- d. Connecting device

Correct Answer: Select (b)
Explanation: Personal fall arrest systems consist of anchorage, full-body harness, and connecting lanyard/device (ABC rule). Slings are material handling rigging gear.

Question 237

Question: There are four different types of machine guards. Which of the following does not belong to the four categories of machine guards?

- a. Fixed
- b. Adjustable
- c. Self-adjusting
- d. Automatic

Correct Answer: Select (d)
Explanation: The four recognized machine guard types are Fixed, Interlocked, Adjustable, and Self-adjusting. "Automatic" is a safety device category, not a guard type.

Question 238

Question: Cranes must meet the design specifications of which of the following organizations?

- a. American Society of Civil Engineers (ASCE)
- b. American National Standard Institute (ANSI)
- c. Consumer Products Safety Commission (CPSC)
- d. National Safety Council (NSC)

Correct Answer: Select (b)
Explanation: All crane design and manufacturing specifications must comply with ANSI (B30 series) standards.

Question 239

Question: The best material for a step ladder used by an electrician who may be working near energized conductors is:

- a. Wood
- b. Fiberglass
- c. Aluminum
- d. Steel

Correct Answer: Select (b)
Explanation: Non-conductive fiberglass is the required material for electrical work (wood can conduct when damp; metals are conductors).

Question 240

Question: Appendix A in 29 CFR 1926 Subpart P dealing with trenching applies when a sloping or benching system is designed as a method of protection for employees from cave-ins. OSHA requires that each soil and rock deposit be classified by a competent person as:

- a. Stable Rock, Type A, Type B, or Type C.
- b. Porous, Solid, Gravel, Sand
- c. Clay, Moist, Solid
- d. Type G, clay, Sand

Correct Answer: Select (a)
Explanation: Soil deposits must be classified by a competent person as Stable Rock, Type A, Type B, or Type C soil.

Question 241

Question: Which of the following statements is true about the Delphi Technique?

- a. It is a structured communication technique.
- b. It is a system safety Technique.
- c. It is a statistical technique for prediction of human error.
- d. It was used by ancient Greeks for design of structural systems.

Correct Answer: Select (a)
Explanation: Delphi technique is a structured communication technique developed as an interactive forecasting method where a panel of experts answers questionnaires anonymously in rounds.

Question 242

Question: Which of the following statements is true about OSHA's Voluntary Protection Programs (VPP)?

- a. The Voluntary Protection Programs (VPP) promote effective worksite-based safety and health.
- b. In the VPP, management, labor, and OSHA establish cooperative relationships at workplaces that have implemented a comprehensive safety and health management system.
- c. Approval into VPP is OSHA's official recognition of the outstanding efforts of employers and employees who have achieved exemplary occupational safety and health.
- d. All the above

Correct Answer: Select (d)
Explanation: All listed statements accurately describe OSHA's Voluntary Protection Programs (VPP).

Question 243

Question: Which of the following statements is true regarding a type II error?

- a. Incorrect rejection of a true null hypothesis.
- b. Correct rejection of a null hypothesis.
- c. Failure to reject a false null hypothesis.
- d. All the above.

Correct Answer: Select (c)
Explanation: A Type II error ($ eta$ error) occurs when statistical testing fails to reject a false null hypothesis.

Question 244

Question: In Globally Harmonized System (GHS), a diamond pictogram featuring a dead tree and dead fish represents:

- a. Toxicity
- b. Explosivity
- c. Flammability
- d. Environmental hazard

Correct Answer: Select (d)
Explanation: The dead tree / dead fish GHS pictogram designates aquatic environmental toxicity hazards.

Question 245

Question: In Globally Harmonized System (GHS), a diamond pictogram featuring a flame over a circle represents:

- a. An oxidizing agent
- b. Flammability
- c. Corrosivity
- d. Explosivity

Correct Answer: Select (a)
Explanation: A flame over a circle designates GHS oxidizing liquids, solids, or gases.

Question 246

Question: OSHA requires inspection of trenches on a _____ basis and as conditions change to ensure elimination of excavation hazards.

- a. Weekly
- b. Daily
- c. Biweekly
- d. Hourly

Correct Answer: Select (b)
Explanation: OSHA 1926 Subpart P requires daily excavation inspections by a competent person prior to the start of work and as conditions change.

Question 247

Question: Ropes used for material handling must be inspected at least once every _____ days.

- a. 7
- b. 10
- c. 30
- d. 14

Correct Answer: Select (c)
Explanation: Material handling ropes and slings must undergo thorough periodic inspections at least every 30 days (or daily visual checks before use).

Question 248

Question: Ropes, chains, and slings should never be used to lift a load if they have which of the following?

- a. Kink
- b. Twist
- c. Knot
- d. Any of the above

Correct Answer: Select (d)
Explanation: Slings/ropes showing kinks, twists, knots, or bird-caging must be removed from service immediately.

Question 249

Question: ISO 45001 deals with which of the following?

- a. Occupational health and safety management systems
- b. Environmental management
- c. Toxic substances control management
- d. Process safety management

Correct Answer: Select (a)
Explanation: ISO 45001 is the global international standard for occupational health and safety management systems.

Question 250

Question: A set of data has a Pearson correlation coefficient of 0.5 between the dependent and independent variables. This indicates that:

- a. As the independent variable increases, the value of the dependent variable decreases.
- b. There is no relationship between the dependent and independent variables.
- c. As the value of the independent variable increases, the value of the dependent variable increases.
- d. As the value of the dependent variable increases, the value of the independent variable decreases.

Correct Answer: Select (c)
Explanation: A positive Pearson correlation ($r = +0.5$) indicates that as the independent variable increases, the dependent variable also increases proportionally.

Question 251

Question: Which of the following is the best description of "safety case approach?"

- a. Uses system safety techniques to identify the hidden system failure modes.
- b. Is a structured argument, supported by a body of evidence, that provides a compelling, comprehensible and valid case that a system is safe for a given application in an operating environment.
- c. Is a methodology that investigates the failure modes of the system by assigning probabilities to each case that can contribute to an accident.
- d. All the above.

Correct Answer: Select (b)
Explanation: A Safety Case is a structured argument, supported by evidence, intended to justify that a system is acceptably safe for a specific application in a specific operating environment.

Question 252

Question: The main elements of a "safety case approach" are "claim," "evidence," "argument" and:

- a. Decision
- b. Support
- c. Inference

- d. Documentation

Correct Answer: Select (c)
Explanation: Safety cases link Claims to Evidence through explicit Arguments using transformed Inference rules.

Question 253

Question: Which of the following statements best describe the Risk Factor Summation (RFS) method?

- a. The RFS-method uses a base-value of a comparable startup for the valuation of your company. This means you compare your startup to other startups and assess whether you have higher or lower risk.
- b. A method that calculates total risk by multiplying probability by severity.
- c. A financial model that sums fixed and variable costs.
- d. An insurance rating model.

Correct Answer: Select (a)
Explanation: The RFS method uses a base value of a comparable startup for business/project valuation by summing adjusted risk factor scores.

Question 254

Question: Which of the following statements is true in regard to business continuity management?

- a. Effective continuity management is the responsibility of every business.
- b. Modern business continuity planning approaches, such as ISO 22301, recognize that organizations cannot operate in isolation.
- c. They depend on an extended network of external suppliers and stakeholders.
- d. All of the above.

Correct Answer: Select (d)
Explanation: All listed statements correctly reflect modern business continuity management principles under standards such as ISO 22301.

Question 255

Question: The six major steps in creating a successful Business Continuity Plan are:

- a. Identify scope, identify key business areas, identify critical functions, identify dependencies, determine acceptable downtime, create plan to maintain operations
- b. Plan, Do, Check, Act, Study, Review
- c. Identify hazards, assess risk, train workers, audit, review, enforce
- d. None of the above

Correct Answer: Select (b)
Explanation: The six major steps in creating a Business Continuity Plan are: (a) Identify scope, (b) Identify key business areas, (c) Identify critical functions, (d) Identify dependencies, (e) Determine acceptable downtime, (f) Create plan to maintain operations.

Question 256

Question: In management literature, gap analysis involves the comparison of actual performance with potential or desired performance. If an organization does not make the best use of current resources, or forgoes investment in capital or technology, it may:

- a. Achieve maximum efficiency
- b. Produce or perform below an idealized potential
- c. Automatically satisfy ISO 9001 standards
- d. Eliminate all operational risks

Correct Answer: Select (b)
Explanation: Failing to optimize resources causes an organization to operate below its idealized potential, revealing a performance gap.

Question 257

Question: In gap analysis, after comparing the current state with the ideal state, the next major steps are:

- a. Identify scope and critical functions
- b. Hire external auditors and retrain staff
- c. File regulatory reports with OSHA
- d. Describe the gap and quantify the difference

Correct Answer: Select (d)
Explanation: Gap analysis requires describing the identified performance gap and quantifying the magnitude of the difference between current and target states.